COMPLETE INTERVIEW EDITION

Deep Learning Handbook — Complete Interview Edition

ANN · CNN · RNN/LSTM/GRU · Transformers · 200+ Interview Questions with Keras Examples

Perceptron to Transformers

Interview-first notes, diagrams, Keras code, mock interview and cheat sheet

Python · scikit-learn

Simple English + Hinglish

Executed code · Verified outputs

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How to Use This Handbook

What you get

A beginner-friendly deep learning handbook for interviews. Seven short chapters explain each topic with an **everyday analogy**, a **simple diagram**, **small formulas with tiny examples** and a **short Keras program explained line by line**. Then come **210 practice questions** (beginner, intermediate, advanced, coding, debugging and scenario) with short answers, Hinglish explanations and follow-up questions, a **mock interview**, a **cheat sheet** and **verified references**.

Hinglish mein samjho

Pehle chapter padho, phir uska code chalao, phir us chapter ke questions ke jawab bol kar do. Har concept pehle roz-marra ki misaal se samjhaya hai, phir chhota formula, phir code. Mushkil words ke saath Hinglish mein chhota explanation hai.

How each chapter is built

Step	What it gives you
1. Everyday analogy	The idea in plain life words
2. Diagram and workflow	The steps as a picture
3. Small formula	One easy formula with a tiny numeric example
4. Comparison table	Quick differences interviewers like to ask
5. Keras code and line-by-line notes	Short code that really ran, with each line explained
6. Uses, pros, cons, mistakes	Real-world examples and traps
7. Summary and practice themes	Last-minute revision

Colour guide

Orange = Hinglish. Pink = analogy. Yellow = line-by-line code notes. Blue = worked example. Red = common mistakes. Purple = summary. Green *Output* panels are the **real output** from TensorFlow 2.21 / Keras 3.15 on small built-in or synthetic data.

Honesty note

All questions are **original practice questions** inspired by themes that public resources (GeeksforGeeks, Naukri Code360, Keras/TensorFlow docs, the original papers) also cover. None is claimed to have been asked at any company, and no frequency numbers are given. The references chapter shows exactly which links were verified.

Answer template for interviews: (1) one-line meaning, (2) how it works in two lines, (3) one tiny example, (4) one limit or trade-off.

PART A

Neural Network Foundations

1. Perceptron and Artificial Neural Networks

2. Activation Functions

3. Loss Functions

4. Optimizers

CHAPTER 1

Perceptron and Artificial Neural Networks

1. The Big Idea in Simple Words

In one line

A **neuron** takes numbers in, multiplies each by a **weight**, adds a **bias**, and passes the result through an **activation**. A **neural network (ANN)** is many neurons in layers. It **learns** by guessing, checking how wrong it was (the **loss**), and fixing its weights a little each time.

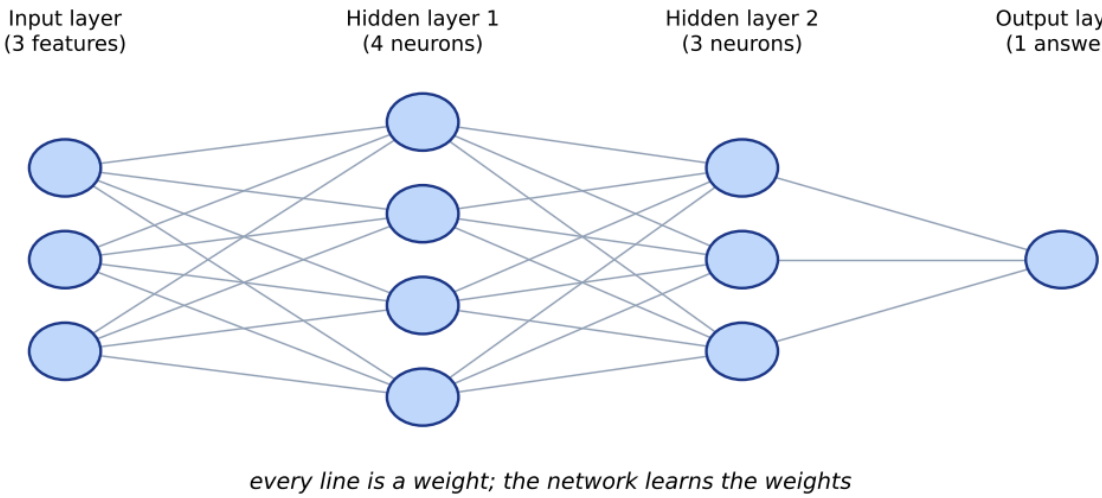
Everyday analogy: learning to cook

You guess how much salt to add (a weight). You taste the dish (the loss tells you how wrong it is). Next time you change the salt a little (an update). After many tries (epochs) the taste is right. The **learning rate** is how big a pinch you change each time.

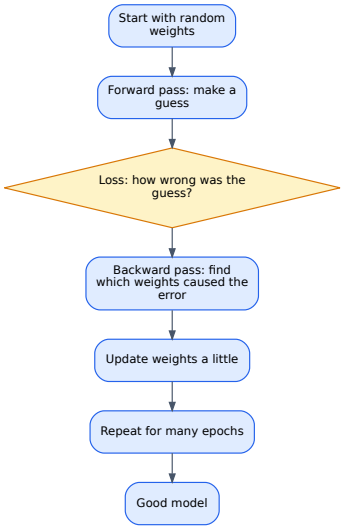
Hinglish mein samjho

Neuron ek chhota calculator hai: inputs ko weights se guna karo, jodo, bias add karo. Network guess karta hai, galti naapta hai (loss), aur weights ko thoda sudhaarta hai. Yahi loop baar-baar chalta hai.

2. Picture: A Small Network



A small neural network. Each circle is a neuron and each line is a weight.



3. Words You Must Know

Word	Simple meaning	Everyday example
Weight	How important an input is	Marks for each subject in a final result
Bias	A starting push, independent of inputs	A free 5 marks added to everyone
Loss	How wrong the guess is	Difference between your guess and the real price
Forward propagation	Input to output, making a guess	Reading a question and answering
Backpropagation	Sending the error back to find who is responsible	Checking which chapter caused wrong answers
Chain rule	Splitting blame along a chain of steps	If A affects B and B affects C, A's effect on C is (A to B) times (B to C)
Learning rate	Size of each fix	Size of a pinch of salt
Epoch	One full pass over all the data	Reading the whole book once
Optimizer	The rule that updates the weights	Your style of correcting mistakes

4. One Small Formula with a Tiny Example

New weight = old weight − learning rate × gradient

Tiny example

Old weight $= 0.50$. The gradient (how much the loss grows if the weight grows) is -0.09. Learning rate $= 0.5$. New weight $= 0.50 - 0.5 \times (-0.09) = 0.545$. The gradient was negative, so the weight went **up**, which lowers the loss.

Hinglish mein samjho

Gradient batata hai weight badhane se loss badhega ya ghatega. Hum hamesha ulti direction mein chalte hain, isliye "minus". Chain rule se gradient nikalte hain: agar A se B aur B se loss badhta hai, to A ka asar = (A ka B par asar) x (B ka loss par asar).

5. Compare: ANN vs CNN vs RNN and Black-box vs White-box

	ANN	CNN	RNN family
Good for	Tables of numbers	Images	Text, speech, time series
Main idea	Layers of neurons	Small filters slide over the image	Remembers the past with a hidden state
Everyday picture	A team voting	A torch scanning a photo	Reading a story line by line
Weakness	Ignores order and shape	Not for word order	Slow; forgets long ago

	Black-box model	White-box model
Examples	Deep networks	Linear regression, small tree
Can you see why it decided?	Hard	Easy
Accuracy on hard data	Often higher	Often lower
Use when	You need accuracy on images or text	Banks or hospitals must explain decisions

6. Code: Train a Small ANN in Keras

PYTHON

```python
import keras
from keras import layers
from sklearn.datasets import load_breast_cancer
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler

X, y = load_breast_cancer(return_X_y=True)
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=0)
scaler = StandardScaler().fit(X_train)
X_train, X_test = scaler.transform(X_train), scaler.transform(X_test)

model = keras.Sequential([layers.Input(shape=(30,)),
                          layers.Dense(16, activation="relu"),
                          layers.Dense(1, activation="sigmoid")])
model.compile(optimizer="adam", loss="binary_crossentropy", metrics=["accuracy"])
model.fit(X_train, y_train, epochs=20, batch_size=32, verbose=0)
print("Test accuracy:", round(model.evaluate(X_test, y_test, verbose=0)[1], 3))
```

OUTPUT

```
Test accuracy: 0.947
```

Line by line, in simple English

1. `import keras` and `layers`: load the deep learning tools. The next three lines load a small medical dataset and tools to split and scale it.
2. `load_breast_cancer`: 569 patients, 30 measurements each; the label says cancer or not.
3. `train_test_split`: keep 20% of patients hidden for the final exam (test data).
4. `StandardScaler().fit(X_train)`: learn the average and spread **from training data only**. Then `transform` puts all numbers on a similar scale, because networks learn better that way.
5. `Sequential([...])`: a stack of layers, one after another. `Input(shape=(30,))` means 30 numbers go in.
6. `Dense(16, activation="relu")`: 16 neurons that each look at all 30 inputs; ReLU keeps positive values and sets negatives to 0.
7. `Dense(1, activation="sigmoid")`: one output neuron giving a number between 0 and 1 (probability of the positive class).
8. `compile`: choose how to learn. `adam` is the optimizer, `binary_crossentropy` is the loss for yes/no answers, and `accuracy` is what we print.
9. `fit(... epochs=20, batch_size=32)`: learn by passing the training data 20 times, 32 patients at a time.
10. `evaluate`: grade the model on the hidden test data and print the accuracy.

7. Uses, Good Points, Limits, Mistakes

- **Real-world uses:** loan risk scores, price prediction, medical risk from lab tests, customer churn.
- **Good:** learns complex patterns without hand-made rules; one method works for many tasks.
- **Limits:** needs lots of data, can memorise (overfit), hard to explain, sensitive to scaling.

Common mistakes

Not scaling the inputs; starting all weights at zero (every neuron then learns the same thing); tuning on the test set; using the wrong loss for the output (see the Loss chapter); forgetting that the test data must stay hidden until the end.

8. Summary

Chapter Summary

- Neuron: inputs × weights + bias, then an activation. Network = layers of neurons.
- Learning loop: forward guess → loss → backward blame (chain rule) → small weight update → repeat for many epochs.
- New weight = old weight − learning rate × gradient.
- **Memory trick:** "Guess, Grade, Blame, Fix" (forward, loss, backprop, update).
- **Practise:** perceptron and XOR, forward vs backward pass, role of the learning rate, why not zero weights, overfitting fixes.

CHAPTER 2

Activation Functions

1. The Big Idea in Simple Words

In one line

An **activation function** is a small rule applied to a neuron's answer. It adds **bending (non-linearity)**, so the network can learn curved, complex patterns and not just straight lines.

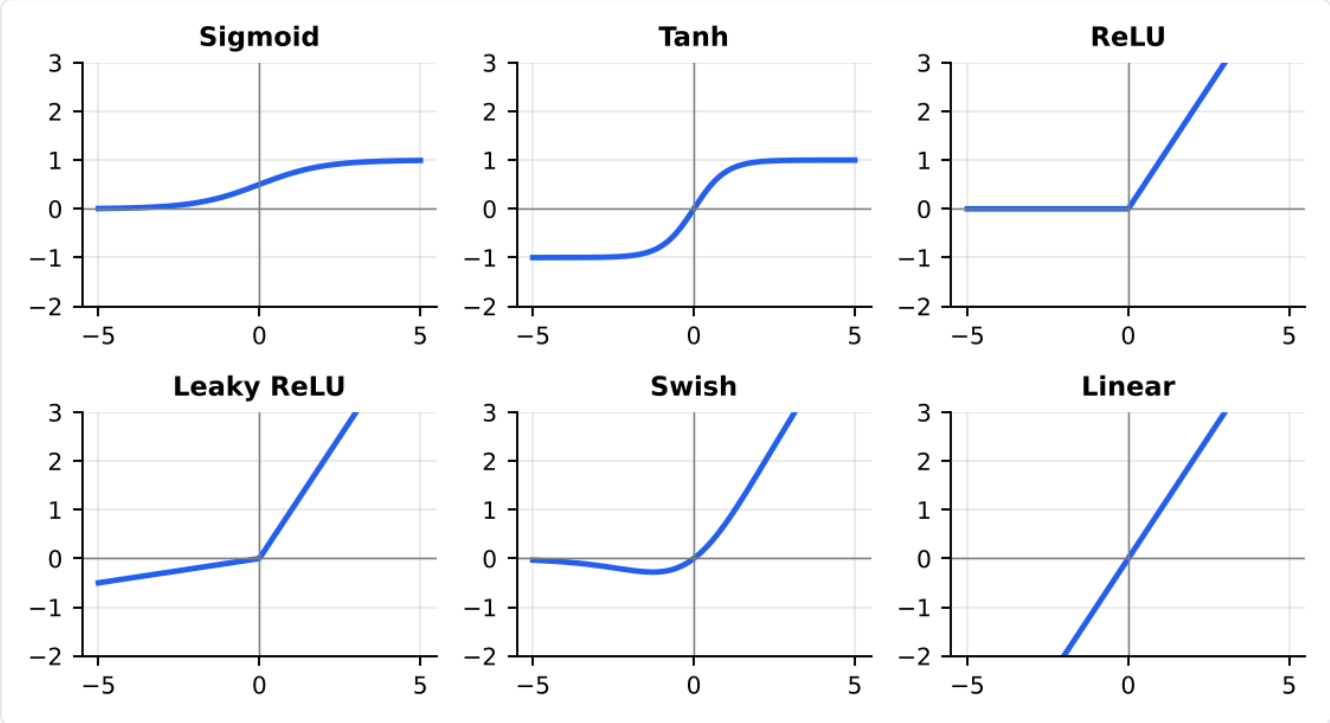
Everyday analogy: a tap or a gate

Think of a **gate** that decides how much water (signal) passes. A plain pipe (linear) passes everything. ReLU is a gate that lets positive water through and blocks negative water. Sigmoid is a gate that squeezes any flow into a number between 0 and 1.

Hinglish mein samjho

Agar activation na ho, to 100 layers bhi sirf ek seedhi line jaisi hongi. Activation network ko mod (bend) deta hai. **Vanishing gradient** matlab: peeche aate-aate seekhne ka signal itna chhota ho jata hai ki shuruaati layers kuch seekhti hi nahi.

2. Picture: The Shapes



Shapes of the main activation functions. Sigmoid squeezes to 0..1, tanh to -1..1, ReLU blocks negatives.



3. The Functions in Plain Words

Function	What it does	Output range	Use it for	Problem
Linear	Passes the value unchanged	Any number	Output of price prediction	No bending
Sigmoid	Squeezes into 0 to 1 (a probability)	0 to 1	Yes/no output	Flat at the ends, so learning stalls
Tanh	Squeezes into -1 to 1	-1 to 1	Inside RNN/LSTM	Also flat at the ends
ReLU	Negative becomes 0, positive stays	0 and above	Default for hidden layers	A neuron can get stuck at 0 ("dying ReLU")
Leaky ReLU	Negative gets a tiny slope (like 0.01)	Any number	Fixes dying ReLU	Slope is fixed by us
PReLU	Like Leaky ReLU, but the slope is **learned**	Any number	When there is enough data	Extra parameters
Swish	A smooth ReLU: $x\times$ sigmoid(x)	Slightly below 0 and up	Deeper networks	A little slower
Softmax	Turns scores into probabilities that add to 1	0 to 1	Output for many classes	Used only at the end

Tiny example: softmax

Scores $[2,\ 1,\ 0.1]$ become probabilities $[0.66,\ 0.24,\ 0.10]$. They add up to 1, and the biggest score gets the biggest share. Think of it as dividing 100 chocolates in proportion to scores.

Tiny example: why gradients vanish

Sigmoid's slope is **at most 0.25**. Passing back through 10 sigmoid layers multiplies by 0.25 ten times: $0.25^{10} \approx 0.000001$. The first layers get almost no signal. ReLU's slope is **1** for positive values, so the signal survives.

4. Which Activation Where?

Situation	Choose
Hidden layers (start here)	ReLU
Many neurons die	Leaky ReLU, PReLU or Swish
Yes/no answer	Sigmoid
One answer out of many classes	Softmax
Many yes/no tags at once	Sigmoid on every output
Predicting a number	Linear

Fixing vanishing gradients: use ReLU, good starting weights (He/Glorot), batch normalisation, residual shortcuts, or LSTM/GRU for sequences. **Exploding gradients:** clip the gradient or lower the learning rate.

5. Code: See the Activations Work

PYTHON

```python
import numpy as np
import keras

scores = np.array([[2.0, 1.0, 0.1]])
print("softmax:", keras.activations.softmax(scores).numpy().round(2))
x = np.array([-2.0, 0.0, 3.0])
print("relu   :", keras.activations.relu(x).numpy())
print("sigmoid:", keras.activations.sigmoid(x).numpy().round(2))
print("tanh   :", keras.activations.tanh(x).numpy().round(2))
```

OUTPUT

```
softmax: [[0.66 0.24 0.1 ]]
relu   : [0. 0. 3.]
sigmoid: [0.12 0.5  0.95]
tanh   : [-0.96  0.    1.  ]
```

Line by line, in simple English

1. `import numpy` and `keras`: numpy holds our small arrays; keras has the activation functions ready-made.
2. `scores = ...`: three class scores in one row (a batch of 1 example).
3. `softmax(scores)`: converts the scores into probabilities. `.numpy().round(2)` turns the result into plain numbers with 2 decimals.
4. `x = ...`: three test inputs: a negative, zero and a positive number.
5. `relu(x)`: -2 becomes 0; 0 and 3 stay the same.
6. `sigmoid(x)`: squeezes each input into 0 to 1 (zero gives exactly 0.5).
7. `tanh(x)`: squeezes each input into -1 to 1 (zero stays 0).

6. Code: Why ReLU Beats Sigmoid in Deep Stacks

PYTHON

```python
import keras, tensorflow as tf
from keras import layers

def first_layer_signal(activation):
    keras.utils.set_random_seed(1)
    model = keras.Sequential([layers.Input(shape=(8,))] +
                             [layers.Dense(16, activation=activation) for _ in range(10)] +
                             [layers.Dense(1)])
    x = tf.random.normal((64, 8), seed=1)
    y = tf.random.normal((64, 1), seed=2)
    with tf.GradientTape() as tape:
        loss = tf.reduce_mean((model(x) - y) ** 2)
    grad = tape.gradient(loss, model.layers[0].kernel)
    return float(tf.norm(grad))

for name in ["sigmoid", "relu"]:
    print(name, "signal reaching the first layer:", f"{first_layer_signal(name):.1e}")
```

OUTPUT

```
sigmoid signal reaching the first layer: 2.2e-07
relu signal reaching the first layer: 3.5e-02
```

Line by line, in simple English

1. `def first_layer_signal(activation)` : a helper that builds a network with the given activation and measures how strong the learning signal is at the **first** layer.
2. `set_random_seed(1)` : makes the run repeatable.
3. The model has **10 hidden layers** of 16 neurons (the list comprehension repeats `Dense` ten times) and one output neuron.
4. `x` and `y` are random inputs and targets, just to compute a loss.
5. `GradientTape` records the calculation so TensorFlow can find gradients (the backprop step).
6. `loss = ...` : average squared error between the model output and `y` .
7. `tape.gradient(loss, first layer weights)` : how much the loss would change for each first-layer weight.
8. `tf.norm(grad)` : one number for the size of that signal; bigger means the layer is learning.
9. The last loop prints the signal for sigmoid and ReLU. Sigmoid's is many times smaller: the vanishing gradient in action.

7. Uses, Mistakes and Summary

Common mistakes

Using sigmoid in many hidden layers; adding softmax inside the model **and** telling the loss `from_logits=True` ; using ReLU as the last layer when the answer can be negative; a very large learning rate that makes many ReLU neurons die.

Chapter Summary

- Activations add bending. Hidden: ReLU. Yes/no: sigmoid. Many classes: softmax. Numbers: linear.
- Sigmoid and tanh flatten at the ends, so gradients vanish in deep stacks; ReLU keeps the signal.
- **Memory trick:** "Sigmoid squeezes, ReLU passes, Softmax shares."
- **Practise:** why activations are needed, dying ReLU, Leaky vs PReLU, softmax vs sigmoid, vanishing vs exploding gradients.

CHAPTER 3

Loss Functions

1. The Big Idea in Simple Words

In one line

A **loss function** gives one number that says **how wrong** the model is. Training tries to make this number small. Numbers (regression) use MSE, MAE or Huber. Classes (classification) use **cross-entropy**.

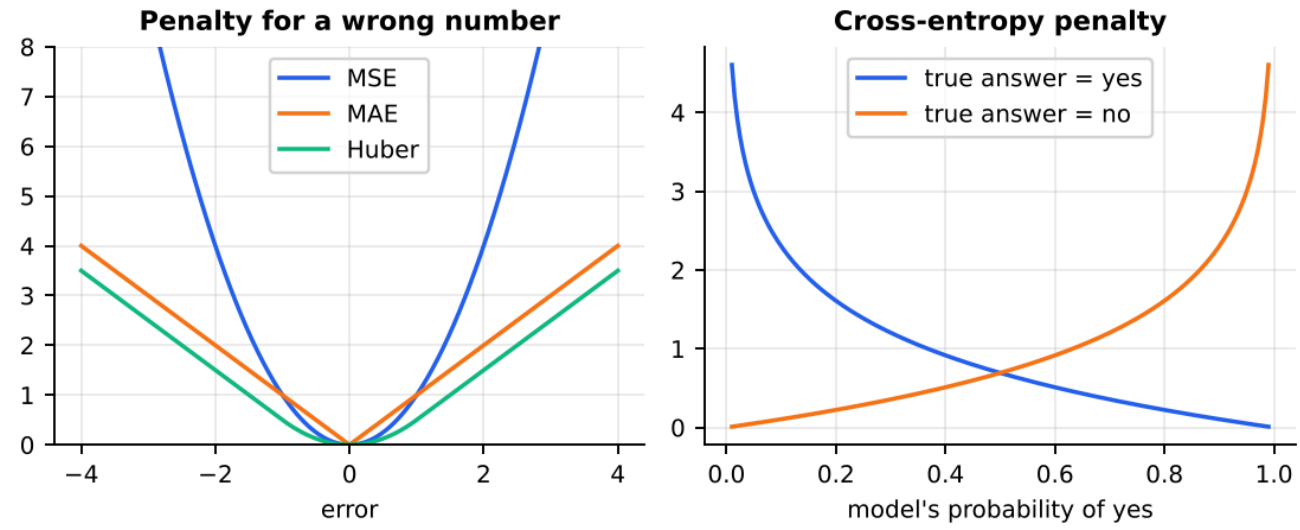
Everyday analogy: marks deducted

A teacher deducts marks for mistakes. **MSE** deducts a lot for big mistakes (square of the error). **MAE** deducts the same amount per unit of error. **Huber** deducts like MSE for small mistakes and like MAE for huge ones. **Cross-entropy** deducts heavily when you are *very sure* but *wrong*.

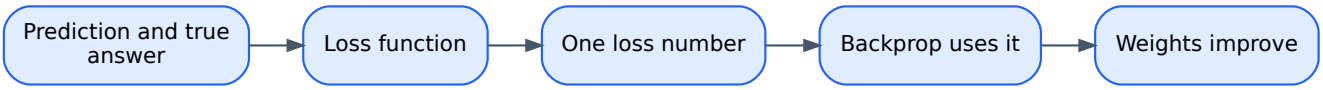
Hinglish mein samjho

Loss model ka "kitne marks kate" number hai. Chhoti galti par thoda, badi galti par zyada. Classification mein sabse zyada saza us answer ko milti hai jisme model bahut confident tha par galat nikla.

2. Picture and Workflow



Left: bigger errors cost much more under MSE. Right: cross-entropy punishes confident wrong answers very strongly.



3. Each Loss in Plain Words, with Tiny Examples

Loss	Meaning	Best when
MSE	Average of (error)²	Clean data; big errors are very bad
MAE	Average of |error|	Data has outliers
Huber	MSE for small errors, MAE for big ones	Outliers, but you want smooth learning
Binary cross-entropy (BCE)	Yes/no answers	Two classes or many independent tags
Categorical cross-entropy	One-hot labels like [0,1,0]	Many classes
Sparse categorical cross-entropy	Integer labels like 1	Many classes, labels are numbers

Tiny example: numbers

True prices $[3, 5, 2]$, predictions $[2.5, 5, 4]$. Errors: $[0.5, 0, -2]$. **MAE** $= (0.5 + 0 + 2)/3 = 0.83$. **MSE** $= (0.25 + 0 + 4)/3 = 1.42$. The one big error (2) dominates MSE. **Huber** (switch point 1) $= 0.54$.

Tiny example: classes

True answer is "yes". If the model says 0.9 for yes the loss is $-\ln 0.9 = 0.11$ (small). If it says 0.1 for yes the loss is $-\ln 0.1 = 2.30$ (big). Three classes with probabilities $[0.2, 0.7, 0.1]$ and the true class is the 2nd: loss $= -\ln 0.7 = 0.36$. **Sparse** and **categorical** give the same number; sparse just takes the label as `1`, and categorical takes `[0,1,0]`.

4. Choose the Right Pair

Task	Last layer	Loss
Yes/no	1 neuron, sigmoid	Binary cross-entropy
One of many, one-hot labels	K neurons, softmax	Categorical cross-entropy
One of many, integer labels	K neurons, softmax	Sparse categorical cross-entropy
Many tags at once	K neurons, sigmoid	Binary cross-entropy
A number	1 neuron, linear	MSE, MAE or Huber

	MSE	MAE	Huber
Big errors	Punished a lot	Punished equally	Punished like MAE
Outliers	Hurt a lot	Little effect	Little effect
Smooth near zero	Yes	No (sharp corner)	Yes

5. Code: Check the Losses by Hand

PYTHON

```python
import numpy as np
import keras

true = np.array([3.0, 5.0, 2.0])
```

```
pred = np.array([2.5, 5.0, 4.0])
print("MSE  :", round(float(keras.losses.MeanSquaredError()(true, pred)), 2))
print("MAE  :", round(float(keras.losses.MeanAbsoluteError()(true, pred)), 2))
print("Huber:", round(float(keras.losses.Huber(delta=1.0)(true, pred)), 2))

probs = np.array([[0.2, 0.7, 0.1]])
print("categorical:", round(float(keras.losses.CategoricalCrossentropy()(np.array([[0, 1, 0.]]), probs)), 2))
print("sparse     :", round(float(keras.losses.SparseCategoricalCrossentropy()(np.array([1]), probs)), 2))
```

OUTPUT

```
MSE  : 1.42
MAE  : 0.83
Huber: 0.54
categorical: 0.36
sparse     : 0.36
```

Line by line, in simple English

1. `true` and `pred` are the real and predicted prices from our tiny example.
2. `MeanSquaredError()(true, pred)` : Keras loss objects are called like functions. This gives the MSE (1.42).
3. `MeanAbsoluteError()` : the MAE (0.83). `Huber(delta=1.0)` : the Huber loss, switching from squared to linear when the error passes 1.
4. `round(float(...), 2)` : turns the result into a normal number with 2 decimals.
5. `probs` : the model's probabilities for 3 classes.
6. `CategoricalCrossentropy` takes the answer as one-hot `[0, 1, 0]` (the 2nd class is correct).
7. `SparseCategoricalCrossentropy` takes the answer as the plain number `1`. Both print 0.36.

6. Uses, Mistakes and Summary

- **Uses:** MSE/MAE for house prices and demand forecasts; cross-entropy for spam, disease and image classes.

Common mistakes

One-hot labels with the *sparse* loss (or the reverse); softmax in the model **plus** `from_logits=True` ; MSE for classification (weak learning signal); comparing loss numbers across different tasks.

Chapter Summary

- Number answers: MSE, MAE, Huber. Class answers: cross-entropy (binary, categorical, sparse).
- Cross-entropy for the true class is $-\ln(\text{probability given to it})$.
- **Memory trick:** "Sparse = integer label, Categorical = one-hot, Binary = yes/no."
- **Practise:** MSE vs MAE vs Huber, why cross-entropy for classes, `from_logits` , imbalanced classes.

CHAPTER 4

Optimizers

1. The Big Idea in Simple Words

In one line

An **optimizer** is the rule that changes the weights to reduce the loss. **Gradient descent** takes small steps downhill. The types differ in how much data they look at before each step, and in how smartly they choose the step.

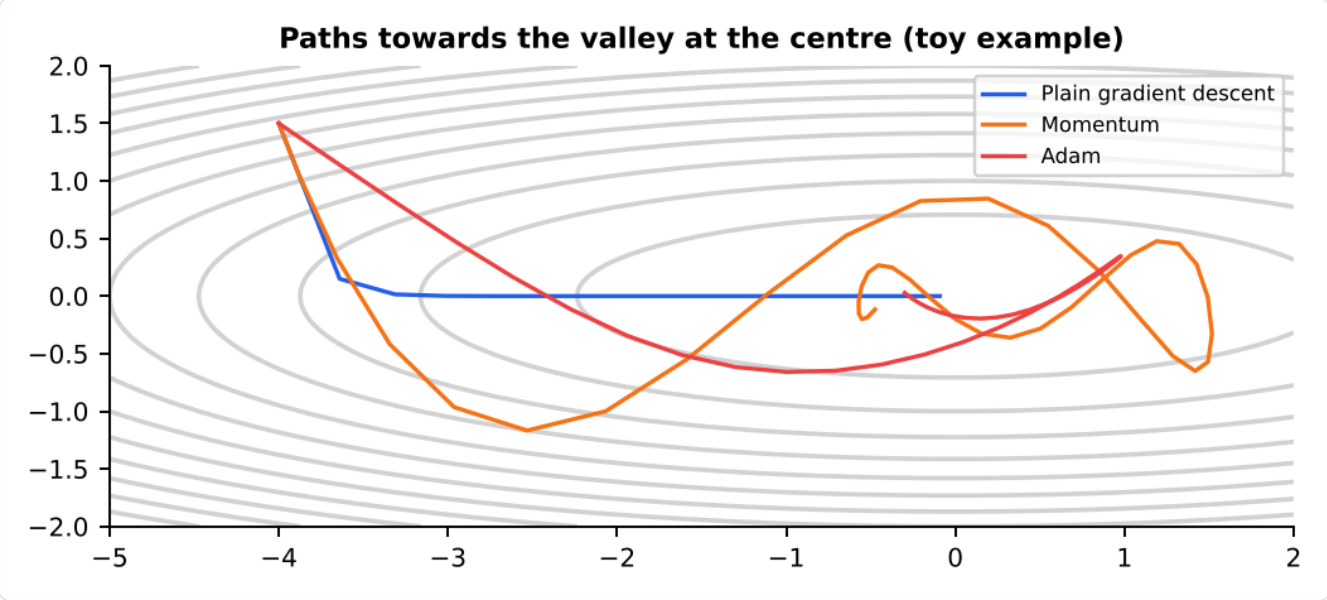
Everyday analogy: walking down a foggy hill

You want to reach the valley but cannot see. You feel the slope under your feet and step downhill. **Batch GD** surveys the whole hill before each step (careful, slow). **Stochastic GD** steps after looking at one spot (fast, shaky). **Mini-batch GD** checks a small patch (balanced). **Momentum** is like a rolling ball that keeps its speed. **AdaGrad** takes smaller steps on terrain you have walked a lot. **Adam** uses speed and smart step sizes together.

Hinglish mein samjho

Optimizer weights ko kaise badalna hai ye tay karta hai. Mini-batch sabse common hai (32, 64 ya 128 samples). Adam sabse popular default hai kyunki wo momentum aur adaptive step size dono use karta hai.

2. Picture and Workflow



A toy valley. Plain gradient descent zig-zags; momentum and Adam reach the centre faster.



3. Types in Plain Words

	Batch GD	Stochastic GD	Mini-batch GD
Data per step	All data	1 sample	A small group (32-128)
Speed	Slow	Fast but shaky	Fast and steady
Memory	High	Low	Medium
Use	Small data	Online learning	Most deep learning

	SGD	Momentum	AdaGrad	Adam
Idea	Step along the slope	Keep previous speed	Smaller steps for often-updated weights	Momentum + smart step sizes
Good point	Simple	Faster, less zig-zag	Good for rare features	Works well with little tuning
Weak point	Slow in narrow valleys	May overshoot	Steps shrink to nearly zero over time	Sometimes a tuned SGD finishes slightly better
Typical learning rate	0.01-0.1	0.01-0.1	0.01-0.1	0.001

Tiny example

The slope (gradient) is always 2 and the learning rate is 0.1. **Plain GD** moves 0.2 every step. **Momentum** (keeps 90% of old speed): step 1 moves 0.2, step 2 moves 0.38, because the speed builds up. **AdaGrad** divides the step by the size of past slopes, so its steps get smaller over time. **Adam** keeps a running average of the slope (direction) and of the slope's size (scale); its first step is about the learning rate (0.1).

4. Code: Compare Optimizers

PYTHON

```python
import numpy as np
import keras
from keras import layers

rng = np.random.RandomState(0)
X = rng.normal(size=(500, 5)).astype("float32")
y = (X @ np.array([[2.0], [-1.0], [0.5], [0.0], [3.0]])).astype("float32")

optimizers = {"SGD": keras.optimizers.SGD(0.01),
              "SGD + momentum": keras.optimizers.SGD(0.01, momentum=0.9),
              "AdaGrad": keras.optimizers.Adagrad(0.1),
              "Adam": keras.optimizers.Adam(0.01)}
for name, opt in optimizers.items():
    keras.utils.set_random_seed(0)
    model = keras.Sequential([layers.Input(shape=(5,)), layers.Dense(8, activation="relu"), layers.Dense(1)])
    model.compile(optimizer=opt, loss="mse")
    history = model.fit(X, y, epochs=15, batch_size=32, verbose=0)
    print(f"{name:15s} final loss = {history.history['loss'][-1]:.4f}")
```

OUTPUT

```
SGD             final loss = 0.1071
SGD + momentum  final loss = 0.0025
AdaGrad         final loss = 0.0485
Adam            final loss = 0.0854
```

Line by line, in simple English

1. `rng` , `X` : make 500 examples with 5 random features each (fake data).
2. `y = X @ [...]` : the true answer is a known mix of the 5 features, so a good model can learn it.
3. `optimizers = {...}` : a list of four optimizers with their learning rates. `momentum=0.9` keeps 90% of the old speed.
4. `for name, opt in ...` : try each optimizer one by one.
5. `set_random_seed(0)` : start from the same random weights each time so the comparison is fair.
6. `Sequential([...])` : a tiny network: 5 inputs, 8 ReLU neurons, 1 output number.
7. `compile(optimizer=opt, loss="mse")` : use this optimizer and the mean squared error loss.
8. `fit(... epochs=15, batch_size=32)` : 15 passes over the data, 32 examples per step (mini-batch).
9. The last line prints the loss after the last epoch. A smaller number means it learned better in the same time.

5. Uses, Mistakes and Summary

- **Where used:** every deep network. Adam for fast starts; SGD with momentum for image models when you can tune it.

Common mistakes

A learning rate that is too large (loss jumps or becomes NaN) or too small (loss barely moves); comparing optimizers for different numbers of epochs; changing the optimizer but keeping the old learning rate; not shuffling data each epoch.

Chapter Summary

- Batch = all data, stochastic = one sample, mini-batch = small group (the usual choice).
- Momentum remembers speed; AdaGrad adapts the step per weight but its steps shrink; Adam combines momentum and adaptive steps (default $\eta = 0.001$).
- **Memory trick:** "Momentum remembers, AdaGrad adapts, Adam does both."
- **Practise:** batch size effects, learning rate schedules, why AdaGrad slows down, Adam vs SGD, gradient clipping.

PART B

Architectures

5. Convolutional Neural Networks (CNN)

6. RNN, LSTM and GRU

7. Transformers

CHAPTER 5

Convolutional Neural Networks (CNN)

1. The Big Idea in Simple Words

In one line

A **CNN** is a network for images. It uses small **filters** that slide over the picture to find patterns such as edges. The same filter is reused everywhere, so it needs far fewer weights than a normal network. Typical order: **Convolution → Pooling → (repeat) → Flatten → Dense → Answer**.

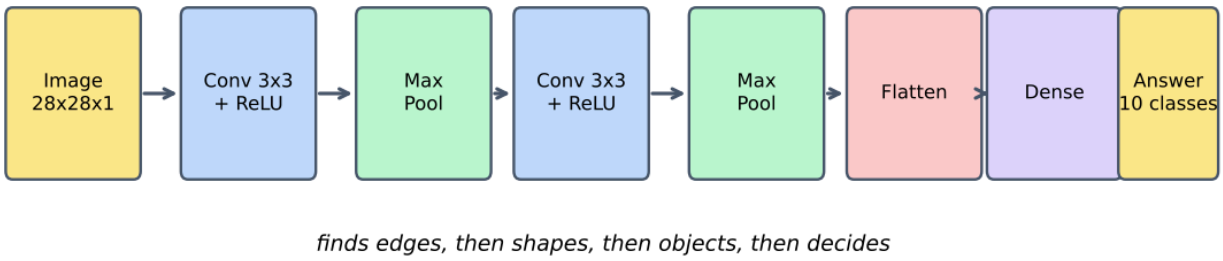
Everyday analogy: a magnifying glass

You scan a big photo with a small magnifying glass. First you spot tiny things (edges), then bigger things (eyes, wheels), and finally you recognise the whole thing (a face or a car). A filter is the magnifying glass; the **same** glass is used on every part of the photo.

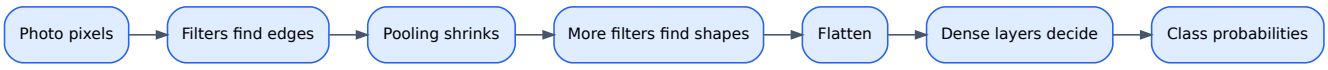
Hinglish mein samjho

Filter ek chhoti window hai (jaise 3x3) jo poori image par ghumti hai aur dekhti hai "yahan edge hai?". Ek hi filter har jagah use hota hai, isliye weights kam. **Pooling** image ko chhota karta hai. **Flatten** feature maps ko ek lambi line bana deta hai taaki Dense layer use kar sake.

2. Picture: A CNN from Photo to Answer

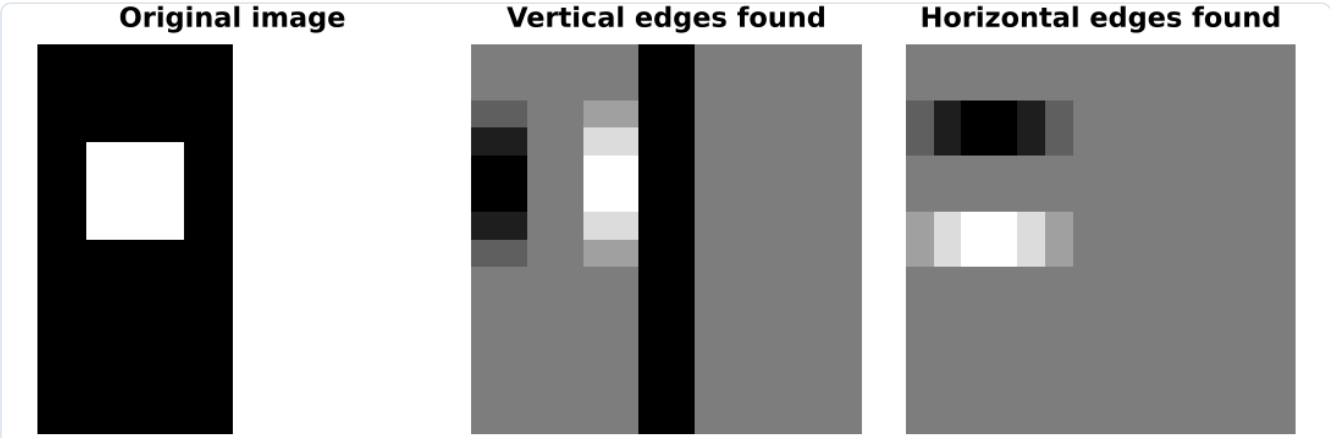


A basic CNN. Convolution finds patterns, pooling shrinks the picture, flatten prepares a list for the final decision.



3. Images, Filters, Padding and Strides

	Grayscale image	Colour (RGB) image
Channels	1	3 (red, green, blue)
Shape	height x width x 1	height x width x 3
Filter depth	1	3
Example	28x28 handwritten digit	224x224 photo



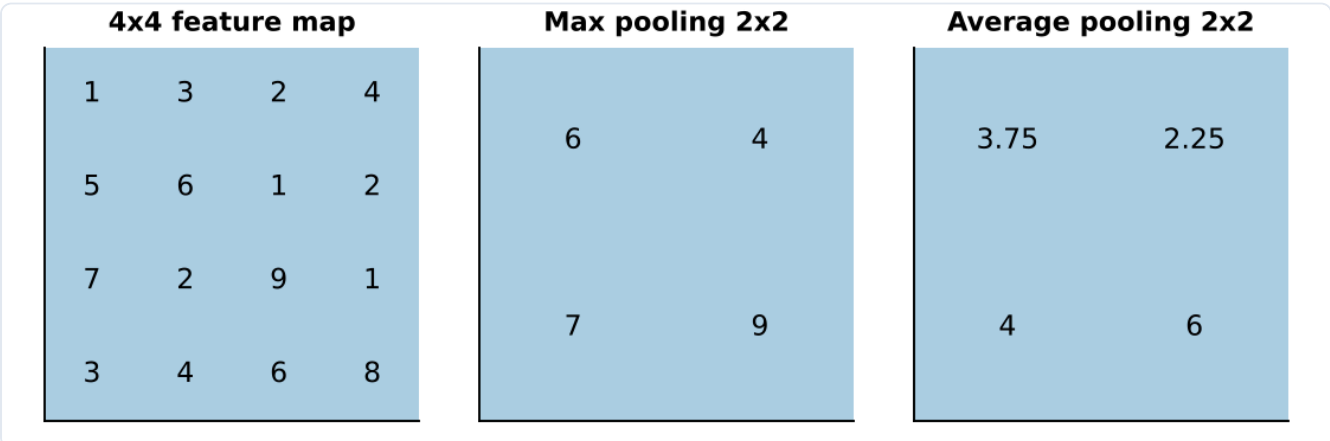
Edge detection: a filter lights up where brightness changes quickly.

- **Filter (kernel):** a small grid of weights that slides across the image. **Edge detection** filters light up where the brightness changes suddenly.
- **Padding:** adding a border of zeros so the picture does not shrink and edge pixels are used fully ("same" keeps the size, "valid" adds nothing).
- **Stride:** how many pixels the filter jumps each step. Bigger stride gives a smaller output.

Tiny example: output size

Rule: **output size = (image size − filter size + 2 × padding) ÷ stride + 1**. Image 5, filter 3, padding 0, stride 1: $(5-3)/1+1=3$. With padding 1: $(5-3+2)/1+1=5$ (same size). With stride 2: $(5-3)/2+1=2$.

4. Pooling and Flattening



Pooling summarises each 2x2 window: max keeps the biggest number, average keeps the mean.

	Max pooling	Average pooling	Min pooling
Keeps	The strongest signal	The overall level	The weakest signal
Good for	Detecting "is a feature present?"	Smooth summaries; **global** average pooling at the end	Rarely used
Everyday picture	Tallest student in each group	Average height in each group	Shortest student in each group

Flatten lays all the final feature-map numbers in one long row so Dense layers can use them. Pooling has **no weights** to learn.

5. Code: A Small CNN for Handwritten Digits

PYTHON

```python
import keras
from keras import layers
from sklearn.datasets import load_digits
from sklearn.model_selection import train_test_split

X, y = load_digits(return_X_y=True)
X = (X / 16.0).reshape(-1, 8, 8, 1)
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=0)

model = keras.Sequential([
    layers.Input(shape=(8, 8, 1)),
    layers.Conv2D(16, 3, padding="same", activation="relu"),
    layers.MaxPooling2D(2),
    layers.Flatten(),
    layers.Dense(10, activation="softmax")])
model.compile(optimizer="adam", loss="sparse_categorical_crossentropy", metrics=["accuracy"])
model.fit(X_train, y_train, epochs=15, batch_size=32, verbose=0)
print("Test accuracy:", round(model.evaluate(X_test, y_test, verbose=0)[1], 3))
print("Parameters:", model.count_params())
```

OUTPUT

```
Test accuracy: 0.961
Parameters: 2730
```

Line by line, in simple English

1. `load_digits` : 1,797 small 8x8 grayscale pictures of digits 0 to 9.
2. `X / 16.0` : pixel values are 0 to 16; dividing makes them 0 to 1, which helps learning.
3. `.reshape(-1, 8, 8, 1)` : give each image the shape height 8, width 8, 1 channel (grayscale), which a CNN expects.
4. `train_test_split` : hide 20% of images for the final test.
5. `Input(shape=(8, 8, 1))` : the network receives 8x8 one-channel images.
6. `Conv2D(16, 3, padding="same", activation="relu")` : 16 different 3x3 filters; "same" keeps the size 8x8; ReLU removes negatives.
7. `MaxPooling2D(2)` : keep the biggest number in every 2x2 block, so 8x8 becomes 4x4.
8. `Flatten()` : turn 4x4x16 numbers into one long list of 256.
9. `Dense(10, activation="softmax")` : 10 outputs, one probability per digit.
10. `compile` : Adam optimizer; **sparse** categorical cross-entropy because labels are plain numbers 0-9.
11. `fit` , `evaluate` : train for 15 passes, then grade on the hidden images and print accuracy and the number of weights.

6. Uses, Good Points, Limits, Mistakes

- **Uses:** photo tagging, face unlock, medical scans, reading number plates, checking factory defects.
- **Good:** fewer weights, finds patterns anywhere in the picture, learns features by itself.
- **Limits:** needs many labelled images, does not understand word order, fooled by changes it never saw (rotation, new lighting) unless you use augmentation.

Common mistakes

Not scaling pixels; wrong shape (forgetting the channel); flattening too early; training on a few images without augmentation; leaving augmentation on when testing.

Chapter Summary

- CNN = filters (find patterns) + pooling (shrink) + flatten + dense (decide).
- Output size = (n − f + 2p) ÷ s + 1. Conv layer weights = (filter × filter × input channels + 1) × number of filters. Pooling has no weights.
- **Memory trick:** "Conv looks, Pool shrinks, Flatten lines up, Dense decides."
- **Practise:** padding vs stride, max vs average pooling, grayscale vs RGB, why CNN has fewer parameters, transfer learning, overfitting on images.

CHAPTER 6

RNN, LSTM and GRU

1. The Big Idea in Simple Words

In one line

These networks read **sequences** (sentences, speech, daily sales) one step at a time and keep a **memory** of what came before. A simple **RNN** keeps one memory (the **hidden state**). **LSTM** adds a protected long-term memory (the **cell state**) controlled by **gates**. **GRU** is a lighter version with fewer gates.

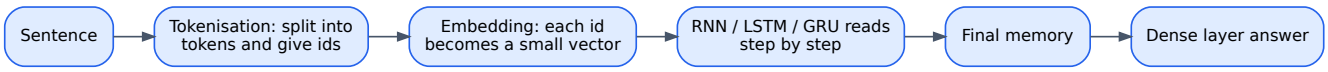
Everyday analogy: reading a story

You read a story sentence by sentence and remember the plot so far. A simple RNN has a **short memory**: after many pages it forgets the start. An LSTM carries a **notebook** (cell state): gates decide what to **erase**, what to **write** and what to **say aloud**.

Hinglish mein samjho

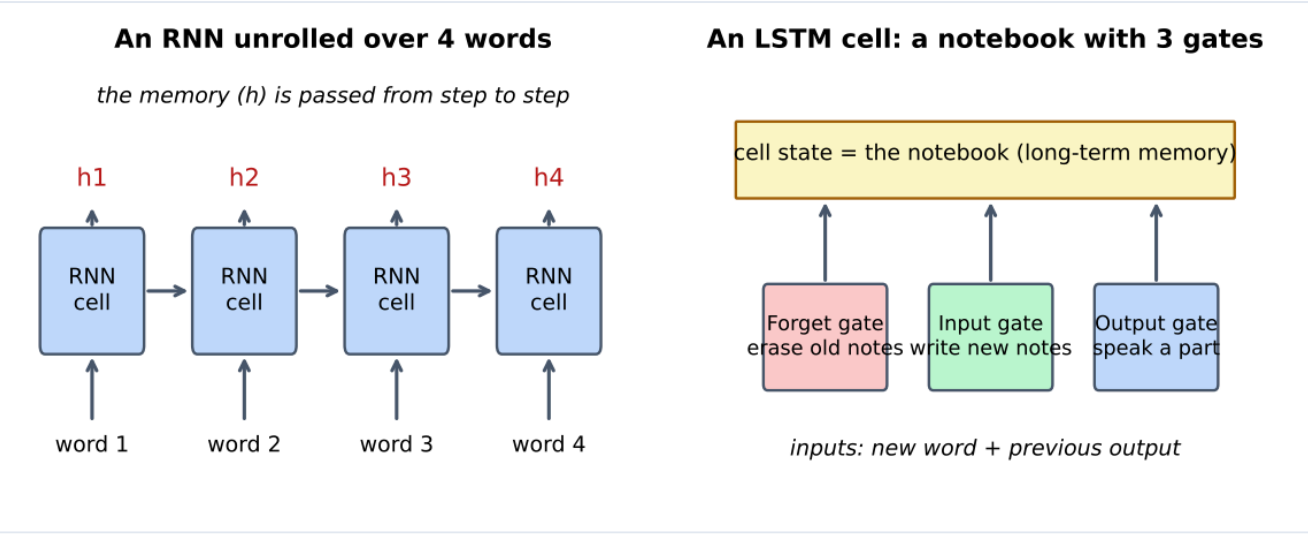
Sequence mein order matter karta hai ("dog bites man" aur "man bites dog" alag hain). RNN har step par pichhli yaadasht (hidden state) saath leke chalta hai. Simple RNN ki yaadasht jaldi kamzor ho jati hai (vanishing gradient); LSTM ka notebook lambi yaadasht sambhalta hai.

2. From Text to Numbers



- **Tokenisation:** cut text into pieces (words or sub-words) and give each piece a number: "i love this film" → [4, 11, 2, 7].
- **Embedding:** each number becomes a short list of learned numbers. Similar words get similar lists, like seating similar students near each other.

3. Picture: The Loop Unrolled and the LSTM Notebook



Left: an RNN reads one word at a time and passes memory forward. Right: an LSTM uses gates to control a notebook.

4. The Gates in Plain Words

Part	Question it answers	Everyday picture
Forget gate (LSTM)	What old information should I erase?	Rubbing out old notes
Input gate (LSTM)	What new information should I write down?	Writing new notes
Output gate (LSTM)	How much of the notebook should I say now?	Reading out a part
Update gate (GRU)	How much to keep old vs take new?	Deciding to update or keep a plan
Reset gate (GRU)	How much of the past to ignore now?	Forgetting the past when the topic changes

	Simple RNN	LSTM	GRU
Memory	One hidden state	Hidden state + cell state	One hidden state
Gates	None	3	2
Long memory	Poor	Good	Good
Weights (relative)	1	About 4	About 3
Speed	Fastest	Slowest	In between
Use	Short sequences	Long sequences	Smaller data, faster training

Tiny example: why simple RNNs forget

At each step the learning signal is multiplied by a number. If that number is 0.9, then after 50 steps $0.9^{50} \approx 0.005$: almost nothing is left of the first word's signal (**vanishing gradient**). If the number is 1.1, then $1.1^{50} \approx 117$: it blows up (**exploding gradient**; fix by clipping). The LSTM's notebook mostly **adds** to its memory instead of multiplying again and again, so the signal survives.

5. Architectures and Limits

Type	Example
Many inputs → one answer	Movie review → positive or negative
One input → many outputs	Photo → caption
Many → many	Translation, tagging each word
Bidirectional	Reads the sentence both forwards and backwards

Limits: must read one step at a time (slow, hard to parallelise); very long sentences are still hard; Transformers now do most text tasks better.

6. Code: Tokenise, Embed and Compare RNN, LSTM and GRU

PYTHON

```python
import numpy as np
import keras
from keras import layers

texts = np.array(["i love this movie", "great and fun", "boring and bad",
                  "i hate this film", "fun great movie", "bad boring film"])
labels = np.array([1, 1, 0, 0, 1, 0])

vectorize = layers.TextVectorization(max_tokens=50, output_sequence_length=5)
vectorize.adapt(texts)
ids = vectorize(texts)
print("ids for first sentence:", ids.numpy()[0])

for name, rnn in [("SimpleRNN", layers.SimpleRNN(8)), ("LSTM", layers.LSTM(8)), ("GRU", layers.GRU(8))]:
    keras.utils.set_random_seed(0)
    model = keras.Sequential([layers.Input(shape=(5,), dtype="int64"),
                              layers.Embedding(50, 8, mask_zero=True), rnn,
                              layers.Dense(1, activation="sigmoid")])
    model.compile("adam", "binary_crossentropy", metrics=["accuracy"])
    model.fit(ids, labels, epochs=60, verbose=0)
    print(name, "weights:", model.count_params(), "| accuracy:", round(model.evaluate(ids, labels, verbose=0)
[1], 2))
```

OUTPUT

```
ids for first sentence: [ 4 11  2  3  0]
SimpleRNN weights: 545 | accuracy: 1.0
LSTM weights: 953 | accuracy: 1.0
GRU weights: 841 | accuracy: 1.0
```

Line by line, in simple English

1. `texts`, `labels`: six tiny movie reviews; 1 means positive, 0 means negative.
2. `TextVectorization(...)`: the **tokeniser**. It will keep up to 50 words and make every sentence exactly 5 tokens long (padding with 0).
3. `adapt(texts)`: learn the vocabulary from our sentences. `vectorize(texts)`: turn each sentence into a row of 5 ids.
4. The loop tries three memory types one by one: SimpleRNN, LSTM and GRU, each with 8 memory units.
5. `Input(shape=(5,), dtype="int64")`: each example is 5 word ids.
6. `Embedding(50, 8, mask_zero=True)`: turn each id into 8 learned numbers; `mask_zero` tells the model to ignore the padding zeros.
7. `rnn`: reads the 5 embedded words in order and ends with one memory vector.
8. `Dense(1, activation="sigmoid")`: one number between 0 and 1: the probability the review is positive.
9. `compile` and `fit`: yes/no loss, Adam optimizer, 60 passes over our six reviews.
10. The print shows the number of weights (LSTM has more than GRU, GRU more than SimpleRNN) and the accuracy. The data is tiny, so this shows how the code works, not real performance.

7. Uses, Mistakes and Summary

- **Uses:** next-word suggestions, speech recognition, stock and sales forecasting, sensor monitoring, sentiment analysis.

Common mistakes

Forgetting `return_sequences=True` when stacking two recurrent layers; not padding or masking sequences; shuffling time-series data and leaking the future into training; not clipping gradients; judging the model only on training sentences.

Chapter Summary

- Tokenise → embed → recurrent layer → dense answer.
- RNN: one memory; LSTM: notebook + 3 gates (forget, input, output); GRU: 2 gates (update, reset).
- Simple RNNs forget because the signal shrinks step after step; LSTM and GRU keep it alive; clip gradients to stop explosions.
- **Memory trick:** "LSTM = Forget, Input, Output; GRU = Update, Reset."
- **Practise:** BPTT, LSTM vs GRU, bidirectional RNNs, many-to-one vs many-to-many, teacher forcing, why Transformers replaced RNNs.

CHAPTER 7

Transformers

1. The Big Idea in Simple Words

In one line

A **Transformer** reads **all words at the same time** and lets every word **look at every other word** to understand its meaning. This looking is called **attention**. Because everything runs in parallel, Transformers train much faster than RNNs.

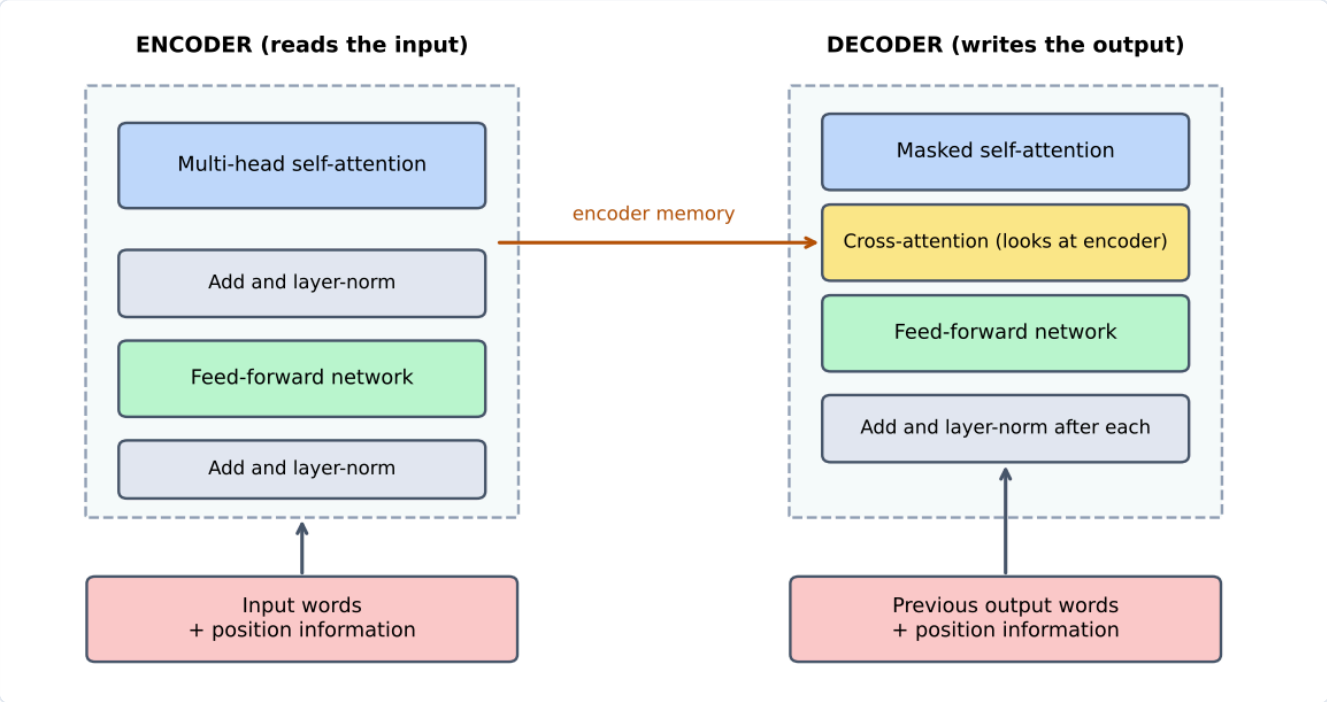
Everyday analogy: a library search

You walk in with a **query** ("I need a book about rivers"). Every book has a **key** (its title tag). You compare your query to all keys; the better the match, the more attention you give. Then you take the **value** (the content) of the best matches. In **self-attention**, every word is both a visitor and a book on the shelf.

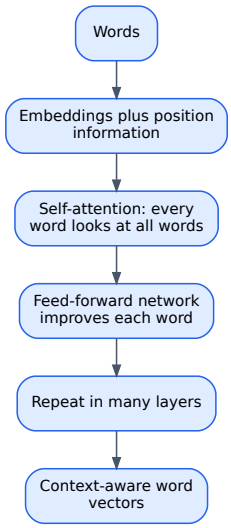
Hinglish mein samjho

"Bank" word river ke saath ho to alag matlab, paise ke saath ho to alag. Self-attention har word ko baaki words dekhne deta hai, isliye wo context samajh leta hai. **Query** = main kya dhundh raha hoon, **Key** = mere paas kya hai, **Value** = asli jaankari.

2. Picture: The Transformer Blocks



A Transformer. The encoder understands the input; the decoder writes the output and also looks at the encoder's memory (cross-attention).



3. Attention Step by Step (No Heavy Maths)

1. Turn each word into three vectors: **Query (Q)**, **Key (K)** and **Value (V)** (three small linear layers).
2. **Score:** compare a word's Q with every K (a dot product). Similar means a high score.
3. **Scale:** divide the scores by a size factor (the square root of the key size) so they do not become too large.
4. **Softmax:** turn the scores into percentages that add to 100%. These are the **attention weights**.
5. **Mix:** the new word vector is the weighted mix of all V vectors.

Short formula: **output = softmax(Q × K / √size) × V**

> **Tiny example**
>
> Two words with vectors $[1, 0]$ and $[0, 1]$, using them directly as Q, K and V. Word 1 scores 1 with itself and 0 with word 2. After scaling and softmax: **67% attention on itself and 33% on the other**. Its new vector is $0.67 \times [1, 0] + 0.33 \times [0, 1] = [0.67,\ 0.33]$. Word 1 now carries a little of word 2.

- **Multi-head attention:** run several attentions side by side (each head looks for a different relation, like grammar or meaning), **concatenate** their outputs, and mix them with one **linear projection**.
- **Positional encoding:** attention alone does not know order, so a pattern of numbers for each position (made of sine and cosine waves) is added to every word.
- **Layer normalisation:** keeps the numbers of each word at a steady scale, so training is stable. **Feed-forward network:** a small two-layer network applied to each word after attention.
- **Contextual embedding:** the final vector of a word depends on its sentence.

4. Training and Testing

> **Everyday analogy: teacher forcing**
>
> While practising, a teacher **shows the correct previous words** so you can learn every step quickly (teacher forcing). In the real exam you have to use **your own earlier words** to write the next one.

	Training	Inference (testing)
Decoder input	The correct previous words (teacher forcing), all at once	Its own previous words, one at a time
Speed	Fast (parallel)	Slower (word by word)
Weights	Updated	Fixed
Dropout	On	Off
Risk	Never practises with its own mistakes (exposure bias)	Mistakes can pile up

5. Comparison Tables

	Encoder-decoder RNN	Transformer
Reads	One word at a time	All words together
Long-distance links	Weak (memory fades)	Direct through attention
Training speed	Slow	Fast on GPUs
Order	Built in	Needs positional encoding
Cost for long text	Grows steadily	Grows with the **square** of the length

	Traditional embeddings (Word2Vec, GloVe)	Contextual embeddings (BERT, GPT)
One word has	One fixed vector	A different vector in every sentence
"bank" (river or money)	Same vector	Different vectors
How made	Look up a table	Run the model on the sentence

6. Code: Attention by Hand, Then a Tiny Transformer

PYTHON

```python
import numpy as np
import keras
from keras import layers

X = np.array([[1.0, 0.0], [0.0, 1.0]])
scores = X @ X.T / np.sqrt(2)
weights = np.exp(scores) / np.exp(scores).sum(axis=1, keepdims=True)
print("attention weights:\n", weights.round(2))
print("new word vectors:\n", (weights @ X).round(2))

mha = layers.MultiHeadAttention(num_heads=2, key_dim=4)
words = np.random.RandomState(0).normal(size=(1, 5, 8)).astype("float32")
out = mha(words, words)
print("output shape:", out.shape)
```

OUTPUT

```
attention weights:
 [[0.67 0.33]
 [0.33 0.67]]
new word vectors:
 [[0.67 0.33]
 [0.33 0.67]]
output shape: (1, 5, 8)
```

Line by line, in simple English

1. `X` : two tiny word vectors. We use them directly as Q, K and V to keep the example small.
2. `X @ X.T` : the dot product of every word with every word (the scores). Dividing by the square root of 2 is the scaling step.
3. `np.exp(scores) / ... .sum(...)` : softmax by hand; every row now adds up to 1.
4. The first print shows the attention weights (0.67 and 0.33). `weights @ X` mixes the vectors: the new word vectors.
5. `MultiHeadAttention(num_heads=2, key_dim=4)` : the ready-made Keras layer with 2 heads, each of size 4.
6. `words` : one sentence of 5 words, each word is 8 numbers.
7. `mha(words, words)` : passing the same input as query and value makes it **self-attention**. The output keeps the shape (1, 5, 8): each word is now context-aware.

PYTHON

```python
import numpy as np
import keras
from keras import layers

keras.utils.set_random_seed(0)
rng = np.random.RandomState(0)
X = rng.randint(1, 20, size=(1500, 12))
y = (X == 7).any(axis=1).astype("float32")
X_train, X_test, y_train, y_test = X[:1200], X[1200:], y[:1200], y[1200:]

inputs = layers.Input(shape=(12,), dtype="int32")
x = layers.Embedding(20, 16)(inputs) + layers.Embedding(12, 16)(keras.ops.arange(12))
x = layers.LayerNormalization()(x + layers.MultiHeadAttention(num_heads=2, key_dim=8)(x, x))
x = layers.LayerNormalization()(x + layers.Dense(16)(layers.Dense(32, activation="relu")(x)))
outputs = layers.Dense(1, activation="sigmoid")(layers.GlobalAveragePooling1D()(x))

model = keras.Model(inputs, outputs)
model.compile("adam", "binary_crossentropy", metrics=["accuracy"])
model.fit(X_train, y_train, epochs=25, batch_size=64, verbose=0)
print("Test accuracy:", round(model.evaluate(X_test, y_test, verbose=0)[1], 2))
```

OUTPUT

```
Test accuracy: 1.0
```

Line by line, in simple English

1. `X` : 1,500 fake sentences of 12 word ids each. `y = 1` if the id 7 appears anywhere, else 0. The model must find this using attention.
2. We keep 1,200 for training and 300 for testing.
3. `Input(shape=(12,))` : each example is 12 ids.
4. First `Embedding` : word ids become 16 numbers each. The second `Embedding` turns positions 0..11 into 16 numbers each (a **learned positional encoding**); we **add** them.
5. `MultiHeadAttention(...)(x, x)` : self-attention. `x + ...` is the **residual shortcut** (keep the original and add the new). `LayerNormalization` steadies the numbers.
6. The next line is the **feed-forward network** (Dense 32 with ReLU, then Dense 16), again with a shortcut and layer normalisation. That is one Transformer encoder block.
7. `GlobalAveragePooling1D` : average all word vectors into one sentence vector. `Dense(1, sigmoid)` : probability that id 7 is present.
8. `compile` , `fit` , `evaluate` : the usual train and test steps. This toy task shows the mechanics, not real-world accuracy.

7. Uses, Limits and Mistakes

- **Uses:** translation, chatbots, search, summarising, code assistants, speech and image models.
- **Limits:** needs lots of data and compute; cost grows with the square of the text length; may produce confident but wrong text.

Common mistakes

Forgetting positional information (the model becomes order-blind); no causal mask in the decoder (it peeks at the future); confusing layer norm with batch norm; treating attention weights as a perfect explanation; mixing up training (teacher forcing) and inference (one word at a time).

Chapter Summary

- Attention = Q asks, K matches, V gives. Output = softmax(Q·K/√size)·V. Multi-head = several attentions, concatenated, then one linear projection.
- Each block: attention + feed-forward, with shortcuts and layer norm; positional encoding gives order; decoder uses a mask and cross-attention.
- Training uses teacher forcing; inference generates word by word.
- **Memory trick:** "Query asks, Key answers who matches, Value is what you take."
- **Practise:** why scale attention, why multi-head, role of the mask, positional encoding, RNN vs Transformer, BERT vs GPT, training vs inference.

PART C

Interview Preparation

8. Interview Q&A · Perceptron and ANN

9. Interview Q&A · Activation Functions

10. Interview Q&A · Loss Functions

11. Interview Q&A · Optimizers

12. Interview Q&A · Training, Regularisation and Initialisation

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CHAPTER 8

Interview Q&A · Perceptron and ANN

Foundations that every deep learning interview starts with. B = beginner, I = intermediate, A = advanced, S = scenario. Each question has a follow-up an interviewer might ask next.

Q1 BEGINNER What is a perceptron?

Interview answer. A single neuron that outputs a step or activation function of a weighted sum of inputs plus a bias, $a = f(\mathbf{w}^\top \mathbf{x} + b)$. It can learn linearly separable problems.

Hinglish samjho. Perceptron ek akela neuron hai: inputs x weights, jodo, bias, phir activation. Sirf seedhi line se alag hone wale data seekh pata hai.

Follow-up. Why can a single perceptron not solve XOR?

Q2 BEGINNER What is an artificial neural network (ANN)?

Interview answer. Layers of connected neurons (input, hidden, output) that learn a mapping from inputs to outputs by adjusting weights to minimise a loss.

Hinglish samjho. Bahut saare neurons layers mein jude hote hain; training se weights aise set hote hain ki loss kam ho.

Follow-up. What does "deep" mean in deep learning?

Q3 BEGINNER How does deep learning differ from classical machine learning?

Interview answer. Deep learning learns features automatically from raw data with many layers, while classical ML usually needs hand-made features; deep models need more data and compute.

Hinglish samjho. Classical ML mein feature engineering hum karte hain, DL mein network khud features seekhta hai, par data aur GPU zyada chahiye.

Follow-up. When would you still prefer gradient boosting over a neural network?

Q4 BEGINNER Why can't XOR be solved by a single perceptron, and how do we fix it?

Interview answer. XOR is not linearly separable, so no single line separates the classes. A hidden layer with a non-linear activation (an MLP) solves it.

Hinglish samjho. XOR ke points ko ek seedhi line se alag nahi kar sakte; ek hidden layer curved boundary bana deti hai.

Follow-up. What is the minimum number of hidden neurons needed for XOR?

Q5 BEGINNER What is forward propagation?

Interview answer. Passing inputs layer by layer, computing $z = Wx + b$ and $a = f(z)$ at each layer, to produce the prediction and then the loss.

Hinglish samjho. Input se output tak ka seedha raasta: har layer weights lagakar activation deti hai, aakhir mein prediction.

Follow-up. What values must you store during the forward pass for backprop?

Q6 BEGINNER What is backpropagation?

Interview answer. An algorithm that computes the gradient of the loss with respect to every weight by applying the chain rule from the output back to the input.

Hinglish samjho. Output ki galti ko peeche ki taraf layer-by-layer baantna ki kis weight ka kitna haath tha.

Follow-up. Is backpropagation the same as gradient descent?

Q7 INTERMEDIATE What is the difference between backpropagation and gradient descent?

Interview answer. Backpropagation *computes* gradients efficiently; gradient descent (or another optimizer) *uses* those gradients to update the weights.

Hinglish samjho. Backprop = gradient nikalna, gradient descent = us gradient se weight badalna. Dono alag kaam hain.

Follow-up. Can you train a network without backpropagation?

Q8 BEGINNER Explain the chain rule in the context of neural networks.

Interview answer. The derivative of a composition is the product of derivatives of its parts: $\frac{\partial L}{\partial w} = \frac{\partial L}{\partial a}\frac{\partial a}{\partial z}\frac{\partial z}{\partial w}$.

Hinglish samjho. Loss weight se kai steps door hai (weight, z, activation, loss), to har step ka derivative multiply karte hain.

Follow-up. Do the derivation for one sigmoid neuron with squared error.

Q9 BEGINNER What are weights and biases?

Interview answer. Weights scale the importance of each input; the bias shifts the neuron's activation threshold. Both are learned parameters.

Hinglish samjho. Weight batata hai input kitna zaroori hai, bias neuron ko aage-peeche shift karta hai.

Follow-up. What happens if you remove all biases?

Q10 BEGINNER What is the learning rate?

Interview answer. The step size multiplier for weight updates, $w \leftarrow w - \eta g$. Too large diverges; too small trains slowly.

Hinglish samjho. Kadam ka size. Bahut bada = loss idhar-udhar uchhalta hai, bahut chhota = bahut slow.

Follow-up. How would you find a good learning rate?

Q11 BEGINNER What is an epoch, a batch and an iteration?

Interview answer. An epoch is one full pass over the training data, a batch is the subset used for one update and an iteration is one update step. Iterations per epoch $= N/\text{batch size}$.

Hinglish samjho. 1000 samples, batch 100 = 10 iterations, aur wo 10 iterations milkar 1 epoch.

Follow-up. How does batch size affect training?

Q12 INTERMEDIATE Why must weights not be initialised to zero?

Interview answer. With zero (or identical) weights every neuron in a layer computes the same output and receives the same gradient, so they never differ (the symmetry problem). Use small random values.

Hinglish samjho. Sab neurons ek jaise rahenge, to network ke 100 neurons bhi ek hi neuron jaisa kaam karenge.

Follow-up. What are Xavier/Glorot and He initialisation and when is each used?

Q13 INTERMEDIATE What is the universal approximation theorem?

Interview answer. A feedforward network with one hidden layer and enough units can approximate any continuous function on a compact set to arbitrary accuracy. It says nothing about how easily it can be learned.

Hinglish samjho. Itna bada hidden layer ho to koi bhi continuous function bana sakte hain, par seekhna aasaan hoga ya nahi, ye alag baat hai.

Follow-up. Then why do we use deep networks instead of one huge hidden layer?

Q14 BEGINNER Why do neural networks need non-linear activation functions?

Interview answer. Without them, a stack of linear layers is equivalent to one linear layer, so the network could only model linear relationships.

Hinglish samjho. Linear ka linear se combination linear hi rehta hai; non-linearity se hi curved patterns seekhte hain.

Follow-up. Prove that two linear layers collapse into one.

Q15 BEGINNER What are the input, hidden and output layers?

Interview answer. The input layer receives features, hidden layers learn intermediate representations and the output layer produces the final prediction in the format the task needs.

Hinglish samjho. Input = data lena, hidden = pattern nikalna, output = answer dena.

Follow-up. How do you choose the number of hidden layers and units?

Q16 INTERMEDIATE How do you choose the number of layers and neurons?

Interview answer. Start small, increase until validation performance stops improving, and use regularisation to control overfitting. There is no formula; tune with validation data.

Hinglish samjho. Chhote network se shuru karo, jab tak validation score sudhar raha hai tab tak bada karo.

Follow-up. What does a very wide but shallow network risk compared with a deep one?

Q17 BEGINNER What is the difference between a parameter and a hyper-parameter in a neural network?

Interview answer. Parameters (weights, biases) are learned during training; hyper-parameters (learning rate, layers, batch size, epochs) are set before training and tuned using validation data.

Hinglish samjho. Parameter model seekhta hai, hyper-parameter hum dete hain.

Follow-up. Name five hyper-parameters you would tune first.

Q18 INTERMEDIATE What is the vanishing gradient problem?

Interview answer. Gradients shrink as they travel back through many layers (often because of sigmoid/tanh saturation and small weights), so early layers learn very slowly.

Hinglish samjho. Peeche aate-aate gradient itna chhota ho jata hai ki shuruaati layers ke weights lagbhag badalte hi nahi.

Follow-up. List four ways to reduce it.

Q19 INTERMEDIATE What is the exploding gradient problem and how do you handle it?

Interview answer. Gradients grow exponentially and cause huge, unstable updates (NaN loss). Use gradient clipping, a smaller learning rate, careful initialisation and normalisation.

Hinglish samjho. Gradient bahut bada ho jata hai aur weights ud jate hain. `clipnorm` se gradient ko ek limit mein rok do.

Follow-up. How would you notice it during training?

Q20 BEGINNER What is a black-box model and how can you explain it?

Interview answer. A model whose internal reasoning is hard to inspect, like a deep network. Explanations come from tools such as feature importance, saliency maps, SHAP and LIME.

Hinglish samjho. Andar kya ho raha hai samajhna mushkil; to bahar se tools lagake dekhte hain kaun sa input zyada zimmedar tha.

Follow-up. When is a white-box model required by law or policy?

Q21 BEGINNER What is a white-box model?

Interview answer. A model whose logic can be read directly, such as linear regression coefficients or a small decision tree.

Hinglish samjho. Jiska formula ya rules seedhe padh sakte hain, jaise logistic regression ya chhota tree.

Follow-up. Can a neural network ever be made interpretable by design?

Q22 INTERMEDIATE What is overfitting in a neural network and how do you prevent it?

Interview answer. The model memorises training data and generalises poorly. Prevent with more data, augmentation, dropout, L2 regularisation, early stopping and a smaller model.

Hinglish samjho. Train accuracy high, validation low. Dropout, early stopping aur data augmentation madad karte hain.

Follow-up. How do you tell overfitting from a data-leak problem?

Q23 INTERMEDIATE What is the difference between a feedforward network and a recurrent network?

Interview answer. A feedforward network passes data in one direction with no memory; a recurrent network feeds its hidden state back, so it can model sequences.

Hinglish samjho. Feedforward mein loop nahi hota, recurrent mein pichhli yaadasht saath chalti hai.

Follow-up. Which would you use for a fixed-size tabular input?

Q24 ADVANCED Why is backpropagation efficient compared with computing each partial derivative separately?

Interview answer. It reuses intermediate results by working backwards through the computation graph (reverse-mode differentiation), so the cost is about two forward passes for all gradients.

Hinglish samjho. Har weight ka gradient alag se nikalne ki jagah peeche ki taraf ek hi baar mein sabka nikal lete hain; common hisse reuse hote hain.

Follow-up. What is the difference between forward-mode and reverse-mode automatic differentiation?

Q25 ADVANCED What is automatic differentiation and how does TensorFlow use it?

Interview answer. It builds a computation graph of operations and applies the chain rule exactly, not by finite differences. `tf.GradientTape` records operations to compute gradients.

Hinglish samjho. Framework har operation yaad rakhta hai aur chain rule khud lagata hai, humein derivative likhna nahi padta.

Follow-up. Why is numerical (finite-difference) gradient checking still useful?

Q26 ADVANCED What is the dying neuron or dead unit problem?

Interview answer. A unit whose pre-activation stays in a region with zero gradient (for ReLU, always negative) outputs zero forever and stops learning, often caused by a large learning rate or poor initialisation.

Hinglish samjho. ReLU ka input hamesha negative rahe to gradient 0, neuron "mar" jata hai.

Follow-up. How do Leaky ReLU and good initialisation help?

Q27 INTERMEDIATE What is the role of the bias neuron in a layer?

Interview answer. It lets the activation shift left or right independent of the inputs, increasing the layer's flexibility.

Hinglish samjho. Bias ke bina line hamesha origin se guzregi; bias use origin se hata sakta hai.

Follow-up. Does batch normalisation make biases redundant?

Q28 ADVANCED Why is the loss surface of a deep network hard to optimise?

Interview answer. It is non-convex with many saddle points, flat regions and sharp ravines; in practice saddle points and poor conditioning matter more than bad local minima.

Hinglish samjho. Surface par bahut gaddhe, mod aur samtal jagah hain; optimizer ko saavdhani se chalna padta hai.

Follow-up. Why do stochastic noise and momentum help here?

Q29 SCENARIO Your network's loss does not decrease at all. How do you debug it?

Interview answer. Check data and labels, scaling, learning rate, loss and output activation pairing, try overfitting a tiny subset, verify the gradient is non-zero, and simplify the model.

Hinglish samjho. Pehle 10-20 samples par model ko overfit karwa ke dekho; agar wo bhi nahi hota to code/loss/labels mein bug hai.

Follow-up. Training works on 20 samples but not on the full set. What next?

Q30 SCENARIO **Training loss goes to NaN after a few steps. What are the likely causes?**

Interview answer. A learning rate that is too high, exploding gradients, log of zero in the loss, division by zero in normalisation, or bad (inf/NaN) input data.

Hinglish samjho. Learning rate kam karo, clipnorm lagao, inputs mein NaN check karo, aur loss mein epsilon/clip lagao.

Follow-up. How do you find which batch caused it?

Q31 SCENARIO **Training accuracy is high but validation accuracy is poor. What do you do?**

Interview answer. It is overfitting: add dropout or weight decay, use augmentation, reduce capacity, stop earlier, gather more data and confirm that train/validation splits are not leaking or mismatched.

Hinglish samjho. Model ratta maar raha hai. Regularisation badhao, data badhao, model chhota karo.

Follow-up. What if validation is *better* than training?

Q32 SCENARIO **When would you choose an ANN over a CNN or an RNN?**

Interview answer. For tabular or fixed-size vector inputs with no spatial or sequential structure.

Hinglish samjho. Image ke liye CNN, sequence ke liye RNN/Transformer, tabular ke liye ANN (ya boosting).

Follow-up. How would you handle tabular data that has categorical features in an ANN?

CHAPTER 9

Interview Q&A · Activation Functions

Know each function's formula, range, derivative behaviour and where it is used.

Q33 BEGINNER What is an activation function and why is it needed?

Interview answer. A function applied to a neuron's weighted sum that adds non-linearity so the network can learn complex patterns.

Hinglish samjho. Activation ke bina network sirf ek seedhi line ke jaisa hota; ye curve banane ki taakat deta hai.

Follow-up. Can you name a network that works without any activation functions?

Q34 BEGINNER Explain the sigmoid function and its main drawbacks.

Interview answer. $\sigma(x) = 1/(1+e^{-x})$ maps to (0,1), good for probabilities. It saturates for large $|x|$ (vanishing gradients) and is not zero-centred.

Hinglish samjho. Output 0 se 1 ke beech, par kinaron par flat ho jata hai to gradient mar jata hai.

Follow-up. Why is its derivative at most 0.25?

Q35 BEGINNER What is the tanh function and how does it compare with sigmoid?

Interview answer. $\tanh$ maps to (-1,1) and is zero-centred, which usually trains better than sigmoid, but it still saturates.

Hinglish samjho. Tanh sigmoid ka zero-centred bhai hai: negative values bhi deta hai, isliye hidden layers mein behtar.

Follow-up. Express tanh in terms of sigmoid.

Q36 BEGINNER What is ReLU and why is it the default for hidden layers?

Interview answer. $\max(0, x)$. It is cheap, does not saturate for positive inputs and gives sparse activations, which speeds up training.

Hinglish samjho. Positive par seedha pass, negative par 0. Simple aur tez, isliye sabse popular.

Follow-up. What is its main weakness?

Q37 INTERMEDIATE What is the dying ReLU problem and how do you fix it?

Interview answer. Units whose input is always negative output 0 and get zero gradient, so they never recover. Use Leaky ReLU/PReLU/ELU, lower learning rates and better initialisation.

Hinglish samjho. Neuron ek baar negative zone mein atak gaya to hamesha band. Leaky ReLU negative side par chhota slope deta hai.

Follow-up. Is ReLU differentiable at 0?

Q38 BEGINNER What is Leaky ReLU?

Interview answer. $f(x) = x$ for $x > 0$ and αx for $x \leq 0$ with a small fixed α like 0.01, so negative inputs still pass a small gradient.

Hinglish samjho. ReLU ka sudhara hua version jo negative side par bhi thoda gradient chhodta hai.

Follow-up. How is PReLU different?

Q39 INTERMEDIATE What is PReLU?

Interview answer. Parametric ReLU: like Leaky ReLU but the negative slope α is a learned parameter.

Hinglish samjho. Slope fix nahi, network khud seekhta hai.

Follow-up. What is the risk of learning α on small datasets?

Q40 INTERMEDIATE What is Swish and why is it used?

Interview answer. $x \cdot \sigma(x)$, a smooth non-monotonic function that often performs slightly better than ReLU in deep networks, at a slightly higher cost.

Hinglish samjho. Smooth curve hai; ReLU ki tarah bada positive side par, aur negative side par thoda dip karke wapas aata hai.

Follow-up. Is Swish bounded above?

Q41 BEGINNER What is the softmax function and where is it used?

Interview answer. $e^{z_i} / \sum_j e^{z_j}$ converts logits into probabilities that sum to 1; used in multi-class output layers.

Hinglish samjho. Saare scores ko probability mein badal deta hai jinka jod 1 ho.

Follow-up. Why do implementations subtract the maximum logit?

Q42 INTERMEDIATE What is the difference between softmax and sigmoid in the output layer?

Interview answer. Softmax gives mutually exclusive class probabilities summing to 1; sigmoid gives an independent probability per class, suitable for binary or multi-label tasks.

Hinglish samjho. Ek hi sahi class ho to softmax; ek image mein kai labels ho sakte hain to sigmoid.

Follow-up. What loss pairs with each?

Q43 BEGINNER Where would you use a linear activation?

Interview answer. In the output layer of regression, where the target can be any real number.

Hinglish samjho. Price predict karna hai to last layer mein koi squash nahi.

Follow-up. Why is a linear activation in hidden layers useless?

Q44 INTERMEDIATE Why is tanh preferred over sigmoid for hidden units?

Interview answer. Its output is zero-centred and its gradient is larger (up to 1), so updates are less biased and training tends to be faster.

Hinglish samjho. Sigmoid ka output hamesha positive hota hai, isse gradient updates ek hi direction mein jhukte hain; tanh mein aisa kam hota hai.

Follow-up. Why do LSTM gates use sigmoid anyway?

Q45 INTERMEDIATE What is the vanishing gradient problem in terms of activation functions?

Interview answer. Saturating activations have derivatives near 0 in their tails, and multiplying many small derivatives across layers makes the gradient vanish.

Hinglish samjho. 0.25 ko 10 baar multiply karoge to lagbhag zero.

Follow-up. Why does ReLU reduce this problem?

Q46 ADVANCED Why is the derivative of sigmoid $\sigma(x)(1-\sigma(x))$?

Interview answer. Differentiating $1/(1+e^{-x})$ gives $e^{-x}/(1+e^{-x})^2$, which equals $\sigma(x)(1-\sigma(x))$, with a maximum of 0.25 at $x=0$.

Hinglish samjho. Ye form chhota hai kyunki forward pass ka σ backward mein reuse ho jata hai.

Follow-up. What is the derivative of tanh?

Q47 ADVANCED **What is the dying-gradient effect with sigmoid outputs and how does cross-entropy help?**

Interview answer. With MSE, saturated sigmoid outputs give tiny gradients even when wrong. Cross-entropy cancels the σ' term so the gradient is $(a - y)$, staying large when the error is large.

Hinglish samjho. Cross-entropy ke saath galat aur confident prediction ka gradient bada rehta hai, isliye sigmoid + BCE accha jodi hai.

Follow-up. Show the cancellation.

Q48 INTERMEDIATE **What is GELU and where is it used?**

Interview answer. A smooth activation $x\,\Phi(x)$ (Gaussian CDF) widely used in Transformer feed-forward layers.

Hinglish samjho. ReLU jaisa par smooth; Transformers mein common.

Follow-up. How does it differ from ReLU near zero?

Q49 SCENARIO **Which output activation and loss do you use for predicting three independent tags per image?**

Interview answer. Three sigmoid units with binary cross-entropy, because tags are not mutually exclusive.

Hinglish samjho. Har tag ka alag haan/na, isliye har output par sigmoid.

Follow-up. How would you set the decision threshold per tag?

Q50 SCENARIO **Your deep sigmoid network does not learn. What changes do you make?**

Interview answer. Switch hidden layers to ReLU-family activations, use He initialisation, add batch normalisation or residual connections and check the learning rate.

Hinglish samjho. Hidden layers mein ReLU lagao aur initialisation sudharo; sigmoid sirf output par rakho.

Follow-up. How would you confirm that vanishing gradients were the cause?

Q51 SCENARIO **Output predictions of a regression model are always positive but targets can be negative. What is wrong?**

Interview answer. The output layer probably uses ReLU or sigmoid. Use a linear output activation.

Hinglish samjho. Last layer mein ReLU negative ko 0 bana deta hai; linear activation lagao.

Follow-up. How would you handle a target that must stay positive?

CHAPTER 10

Interview Q&A · Loss Functions

Match the loss to the task and the label format, and explain the gradient behaviour.

Q52 BEGINNER What is a loss function?

Interview answer. A function that measures how far predictions are from targets; training minimises it.

Hinglish samjho. Model ki galti ko ek number mein badalna.

Follow-up. What is the difference between a loss and a metric?

Q53 BEGINNER What are MSE and MAE?

Interview answer. MSE is the mean of squared errors; MAE is the mean of absolute errors.

Hinglish samjho. MSE badi galti ko zyada saza deta hai, MAE sab ko barabar.

Follow-up. Which is more sensitive to outliers?

Q54 BEGINNER When should you use MAE instead of MSE?

Interview answer. When the data has outliers or when every unit of error costs the same.

Hinglish samjho. Outlier se bachna ho to MAE.

Follow-up. What is the drawback of MAE's gradient?

Q55 INTERMEDIATE What is Huber loss and why is it useful?

Interview answer. It is quadratic for small errors and linear for large errors (threshold δ), combining MSE's smooth gradient with MAE's robustness.

Hinglish samjho. Chhoti galti par MSE, badi galti par MAE jaisa; δ batata hai kahan switch karna hai.

Follow-up. How would you choose δ?

Q56 BEGINNER What is binary cross-entropy?

Interview answer. $-[y \ln p + (1-y)\ln(1-p)]$, the loss for two-class or multi-label problems with sigmoid outputs.

Hinglish samjho. Sahi class ki probability ka $-\ln$; confident aur galat par bahut bada.

Follow-up. What is BCE for $y = 1$ and $p = 0.1$?

Q57 BEGINNER What is categorical cross-entropy?

Interview answer. $-\sum_k y_k \ln p_k$ for one-hot targets and softmax outputs; it reduces to $-\ln p_{\text{true class}}$.

Hinglish samjho. Sirf sahi class ki probability ka $-\ln$ bachta hai.

Follow-up. Why does it ignore the other classes?

Q58 BEGINNER What is sparse categorical cross-entropy?

Interview answer. The same loss as categorical cross-entropy but it takes integer class labels instead of one-hot vectors.

Hinglish samjho. Label 3 seedha de do, one-hot [0,0,0,1] banane ki zaroorat nahi.

Follow-up. When is it more convenient or memory efficient?

Q59 INTERMEDIATE What does `from_logits=True` mean?

Interview answer. The model outputs raw logits and the loss applies softmax/sigmoid internally, which is more numerically stable.

Hinglish samjho. Last layer mein softmax nahi, loss khud lagata hai. Dono jagah lagaoge to galat hoga.

Follow-up. What goes wrong if the model already applies softmax and you also set `from_logits=True` ?

Q60 INTERMEDIATE Why is cross-entropy preferred over MSE for classification?

Interview answer. With sigmoid/softmax outputs, MSE gives weak gradients when predictions are confidently wrong, while cross-entropy gives strong gradients and matches maximum likelihood.

Hinglish samjho. Cross-entropy galat par bada gradient deta hai; MSE saturate ho jata hai.

Follow-up. Link cross-entropy to maximum likelihood.

Q61 INTERMEDIATE How do you choose a loss for a regression problem with outliers?

Interview answer. Use MAE or Huber, or transform the target (for example log), and evaluate with a robust metric.

Hinglish samjho. Outliers hain to Huber ya MAE.

Follow-up. What if errors should be relative to the target size?

Q62 ADVANCED What is the relationship between cross-entropy and KL divergence?

Interview answer. Cross-entropy $H(p, q) = H(p) + D_{KL}(p \| q)$; with fixed labels $H(p)$ is constant, so minimising cross-entropy minimises KL divergence.

Hinglish samjho. Model ka distribution asli distribution se kitna door hai, wahi measure ho raha hai.

Follow-up. What is label smoothing and how does it change the target?

Q63 ADVANCED What is focal loss and when is it used?

Interview answer. A cross-entropy variant that down-weights easy examples with a factor $(1 - p_t)^{\gamma}$, helping with heavy class imbalance such as in object detection.

Hinglish samjho. Aasaan examples ka weight kam, mushkil examples par focus.

Follow-up. Compare it with class weighting.

Q64 INTERMEDIATE How do you handle class imbalance in the loss?

Interview answer. Use class weights, focal loss, resampling or threshold tuning, and evaluate with precision, recall and PR-AUC.

Hinglish samjho. Chhoti class ki galti ko zyada weight do.

Follow-up. Why is accuracy misleading here?

Q65 BEGINNER What loss and output layer do you use for binary, multi-class and regression tasks?

Interview answer. Binary: sigmoid + BCE. Multi-class: softmax + (sparse) categorical cross-entropy. Regression: linear + MSE/MAE/Huber.

Hinglish samjho. Task ke hisaab se last layer aur loss ki jodi yaad rakho.

Follow-up. What changes for a multi-label task?

Q66 INTERMEDIATE Can a loss function be non-differentiable?

Interview answer. The training loss must be (sub)differentiable for gradient descent; non-differentiable metrics like accuracy cannot be optimised directly, so we optimise a smooth surrogate such as cross-entropy.

Hinglish samjho. Accuracy ka gradient nahi hota, isliye cross-entropy ko proxy banate hain.

Follow-up. Name another surrogate loss.

Q67 ADVANCED **Why does MSE assume Gaussian noise?**

Interview answer. Minimising squared error equals maximum likelihood when targets have independent Gaussian noise; MAE corresponds to Laplace noise.

Hinglish samjho. Squared error ke peeche Gaussian noise ka assumption chhupa hai.

Follow-up. What does that imply when noise is heavy-tailed?

Q68 SCENARIO **Your classifier outputs 0.5 for everything and loss stays near 0.693. What is wrong?**

Interview answer. $\ln 2 = 0.693$ is the BCE of a model that outputs 0.5, so it is not learning: check the learning rate, labels, data scaling, dead units or a missing activation.

Hinglish samjho. Loss 0.693 matlab model sirf anuman laga raha hai (random guess). Pehle labels aur learning rate check karo.

Follow-up. What loss value would you expect for random guessing with 10 classes?

Q69 SCENARIO **One batch has a very large loss spike every few epochs. What are possible causes?**

Interview answer. Outliers or mislabelled samples, too high a learning rate, unstable normalisation statistics or exploding gradients; use robust losses, clipping and data inspection.

Hinglish samjho. Koi kharab sample ya bada learning rate; gradient clipping aur data check karo.

Follow-up. How would you find the offending samples?

CHAPTER 11

Interview Q&A · Optimizers

Be ready to write the update rule, explain what each optimizer fixes and give a practical default.

Q70 BEGINNER **What is gradient descent?**

Interview answer. An iterative method that updates parameters in the direction opposite to the gradient of the loss: $w \leftarrow w - \eta \nabla L$.

Hinglish samjho. Pahaad se utarne ki tarah: dhalan ki ulti direction mein kadam.

Follow-up. What guarantees convergence for convex problems?

Q71 BEGINNER **What is the difference between batch, stochastic and mini-batch gradient descent?**

Interview answer. Batch uses all samples per update, stochastic uses one and mini-batch uses a small group (e.g. 32-256); mini-batch balances speed, memory and gradient noise.

Hinglish samjho. Batch = sabko dekhkar kadam, SGD = ek ko dekhkar, mini-batch = thode logon ko dekhkar.

Follow-up. Why do GPUs favour mini-batches?

Q72 BEGINNER **What are the advantages of SGD over batch gradient descent?**

Interview answer. Cheaper updates, works on huge datasets and its noise can help escape shallow minima and saddle points.

Hinglish samjho. Har update tez hai aur thoda shor optimizer ko atakne se bachata hai.

Follow-up. What is the disadvantage of its noise near the minimum?

Q73 INTERMEDIATE **How does batch size affect training?**

Interview answer. Small batches give noisier gradients, more updates and often better generalisation; large batches are faster per epoch and stable but may need a higher learning rate and generalise worse.

Hinglish samjho. Chhota batch = shor aur regularisation jaisa effect; bada batch = smooth par kabhi kabhi sharp minima.

Follow-up. What is the linear-scaling rule for the learning rate?

Q74 BEGINNER What is momentum in optimisation?

Interview answer. It keeps an exponentially decaying sum of past gradients (velocity) so updates accelerate along consistent directions and damp oscillations.

Hinglish samjho. Bhaari gend ki tarah, pichhli speed yaad rakhta hai.

Follow-up. What does $\beta = 0.9$ mean?

Q75 INTERMEDIATE What is Nesterov momentum?

Interview answer. A momentum variant that computes the gradient at the look-ahead position, giving a slightly better correction.

Hinglish samjho. Pehle aage dekho, phir gradient lo.

Follow-up. How do you enable it in Keras?

Q76 BEGINNER What problem does AdaGrad address?

Interview answer. It gives each parameter its own learning rate, scaled by the inverse root of its accumulated squared gradients, so rarely updated (sparse) features get larger steps.

Hinglish samjho. Jin weights ko kam update mila unka learning rate bada, jinko zyada mila unka chhota.

Follow-up. What is AdaGrad's weakness?

Q77 INTERMEDIATE Why does AdaGrad's learning rate decay too much?

Interview answer. The accumulated sum of squared gradients only grows, so the effective step shrinks towards zero and learning stalls.

Hinglish samjho. Sum badhta hi jata hai, isliye kadam chhota hota jata hai. RMSProp isse purane gradients ko bhulakar sudharta hai.

Follow-up. How does RMSProp fix this?

Q78 BEGINNER What is the Adam optimizer?

Interview answer. Adam combines momentum (first-moment average) with RMS-based adaptive scaling (second moment) plus bias correction; defaults $\beta_1 = 0.9$, $\beta_2 = 0.999$, $\eta = 10^{-3}$.

Hinglish samjho. Momentum + adaptive learning rate dono ek saath; isliye default choice.

Follow-up. Why is bias correction needed in the first steps?

Q79 INTERMEDIATE Write the Adam update equations.

Interview answer. $m = \beta_1 m + (1-\beta_1) g$, $v = \beta_2 v + (1-\beta_2) g^2$, $\hat{m} = m/(1-\beta_1^t)$, $\hat{v} = v/(1-\beta_2^t)$, $w \leftarrow w - \eta \hat{m}/(\sqrt{\hat{v}} + \epsilon)$.

Hinglish samjho. Pehla moment direction batata hai, doosra moment scale; dono se step banta hai.

Follow-up. What is the size of the very first Adam step?

Q80 INTERMEDIATE When might SGD with momentum beat Adam?

Interview answer. Often in image tasks with a well-tuned schedule it reaches slightly better final generalisation, while Adam converges faster and needs less tuning.

Hinglish samjho. Adam jaldi seekhta hai, par tuned SGD kabhi kabhi final accuracy mein aage nikalta hai.

Follow-up. What is AdamW and why does it matter?

Q81 INTERMEDIATE What is a learning-rate schedule?

Interview answer. A rule to change the learning rate during training (step decay, cosine, warm-up, reduce-on-plateau) to speed up early progress and fine-tune later.

Hinglish samjho. Shuru mein bade kadam, baad mein chhote kadam.

Follow-up. Why do Transformers use warm-up?

Q82 INTERMEDIATE What is learning rate warm-up?

Interview answer. Starting with a very small rate and increasing it over the first steps to avoid unstable early updates, especially for adaptive optimizers and Transformers.

Hinglish samjho. Shuru mein sab estimates kachche hote hain, isliye dheere se speed badhate hain.

Follow-up. What happens without warm-up in large Transformer training?

Q83 INTERMEDIATE What are saddle points and local minima, and which matters more in deep networks?

Interview answer. A saddle point has zero gradient but curves up in some directions and down in others. In high dimensions saddle points and flat regions are more common obstacles than bad local minima.

Hinglish samjho. Ghode ki zeen jaisi jagah jahan gradient 0 hai par minimum nahi.

Follow-up. How does momentum or noise help?

Q84 ADVANCED Why does Adam sometimes generalise worse than SGD?

Interview answer. Adaptive scaling can converge to sharper minima and its weight decay interacts with the adaptive step when implemented as L2; decoupled weight decay (AdamW) and tuning reduce the gap.

Hinglish samjho. Adam ka adaptive step kabhi kabhi sharp minimum par le jata hai. AdamW weight decay ko alag rakhta hai.

Follow-up. What is the difference between L2 regularisation and weight decay in Adam?

Q85 ADVANCED What is gradient clipping and when do you use it?

Interview answer. Rescaling gradients when their norm exceeds a threshold to prevent exploding updates, common in RNNs and Transformers (`clipnorm` or `clipvalue`).

Hinglish samjho. Gradient bahut bada ho to use ek limit tak kaat do.

Follow-up. Clip by value vs by norm?

Q86 INTERMEDIATE What does "convergence" mean and how do you detect it?

Interview answer. The loss and validation metric stop improving. Monitor curves and use early stopping or reduce-on-plateau callbacks.

Hinglish samjho. Jab loss badalna band kar de to model converge hua.

Follow-up. What if training loss plateaus high?

Q87 SCENARIO Your loss oscillates wildly between epochs. What do you try?

Interview answer. Lower the learning rate, increase batch size, add momentum or an adaptive optimizer, use gradient clipping and check the data pipeline.

Hinglish samjho. Kadam bada hai; learning rate kam karo ya schedule lagao.

Follow-up. How could you find the maximum stable learning rate quickly?

Q88 SCENARIO Loss decreases extremely slowly. What do you check?

Interview answer. Learning rate too low, poor initialisation, unscaled inputs, vanishing gradients or too strong regularisation; try a higher rate, normalise inputs and use Adam.

Hinglish samjho. Learning rate bahut chhota ho sakta hai ya inputs scale nahi hue.

Follow-up. What does a learning-rate range test do?

Q89 SCENARIO **Which optimizer and settings would you start with for a new CNN or Transformer project?**

Interview answer. Adam or AdamW with $\eta \approx 10^{-3}$ (CNN) or 10^{-4} (Transformer), warm-up and decay, then compare with SGD+momentum if you have time to tune.

Hinglish samjho. Pehle Adam default se baseline banao, phir zaroorat ho to SGD try karo.

Follow-up. How would you justify switching optimizers?

Q90 INTERMEDIATE **What is the difference between an optimizer and a loss function?**

Interview answer. The loss defines what to minimise; the optimizer defines how to move the weights to minimise it.

Hinglish samjho. Loss manzil hai, optimizer raasta.

Follow-up. Can two optimizers reach different final solutions on the same loss?

CHAPTER 12

Interview Q&A · Training, Regularisation and Initialisation

Practical topics that decide whether a network actually trains well.

Q91 BEGINNER What is dropout?

Interview answer. During training it randomly sets a fraction of activations to zero, preventing co-adaptation of neurons; at inference all units are used.

Hinglish samjho. Training mein kuch neurons ko randomly band kar dete hain taaki network kisi ek neuron par nirbhar na ho.

Follow-up. How is the output scaled at test time?

Q92 BEGINNER What is early stopping?

Interview answer. Stop training when validation loss stops improving and restore the best weights.

Hinglish samjho. Jab validation score girne lage, wahin ruk jao.

Follow-up. How do you choose the patience value?

Q93 INTERMEDIATE What are L1 and L2 regularisation in neural networks?

Interview answer. They add a penalty on weights to the loss: L1 ($\lambda \sum |w|$) encourages sparse weights; L2 ($\lambda \sum w^2$, weight decay) shrinks weights smoothly.

Hinglish samjho. Weights ko bade hone se rokna; L1 kai weights ko zero kar deta hai, L2 sabko chhota rakhta hai.

Follow-up. Why does a smaller weight norm help generalisation?

Q94 BEGINNER What is data augmentation?

Interview answer. Creating modified copies of training data (flips, crops, noise) to increase diversity and reduce overfitting.

Hinglish samjho. Ek hi photo ko ghuma-fira kar nayi photo banana.

Follow-up. When is augmentation harmful?

Q95 INTERMEDIATE What is batch normalisation and why does it help?

Interview answer. It normalises layer activations over the mini-batch and applies learned scale and shift, which stabilises and speeds training and allows larger learning rates.

Hinglish samjho. Har layer ka input ek jaisa scale mein rakhta hai, to training stable hoti hai.

Follow-up. How does it behave differently in training and inference?

Q96 INTERMEDIATE What is the difference between batch normalisation and layer normalisation?

Interview answer. Batch norm normalises across the batch for each feature; layer norm normalises across the features of each sample, so it works with small batches and sequences (used in Transformers).

Hinglish samjho. Batch norm batch dekhta hai, layer norm ek sample ke andar dekhta hai.

Follow-up. Why is batch norm awkward for RNNs?

Q97 INTERMEDIATE What are Xavier (Glorot) and He initialisation?

Interview answer. Variance-scaling initialisers: Xavier suits tanh/sigmoid ($\mathrm{Var} = 2/(n_{in} + n_{out})$); He suits ReLU ($\mathrm{Var} = 2/n_{in}$). They keep signal and gradient scales stable across layers.

Hinglish samjho. Sahi shuruaati weights se activations na bahut bade na bahut chhote rehte hain.

Follow-up. Why does ReLU need a larger variance?

Q98 INTERMEDIATE What is transfer learning?

Interview answer. Reusing a model trained on a large dataset and fine-tuning it on a smaller related task, saving data and compute.

Hinglish samjho. Pehle se seekha hua model leke apne kaam ke liye thoda sa aur train karna.

Follow-up. When do you freeze layers vs fine-tune everything?

Q99 INTERMEDIATE What are the training, validation and test sets used for?

Interview answer. Training fits weights, validation tunes hyper-parameters and stops early, and the test set gives one final unbiased estimate.

Hinglish samjho. Test set ko tuning mein mat chhuo.

Follow-up. What is data leakage in deep learning projects?

Q100 ADVANCED What is the bias-variance trade-off for deep networks and what is "double descent"?

Interview answer. Classical theory says more capacity raises variance; modern over-parameterised networks can generalise better again past the interpolation point (double descent), helped by implicit regularisation.

Hinglish samjho. Bahut bade models bhi kabhi kabhi achha generalise karte hain; ye classical rule ka apvaad hai.

Follow-up. Does that mean you should skip regularisation?

Q101 INTERMEDIATE How do you diagnose a model using learning curves?

Interview answer. Large gap between train and validation means overfitting; both high loss means underfitting; noisy curves suggest a high learning rate or small batches.

Hinglish samjho. Dono curves dekho: gap = overfit, dono upar = underfit.

Follow-up. What would you change for each case?

Q102 SCENARIO You have only 500 labelled images. How do you build a classifier?

Interview answer. Use transfer learning from a pretrained CNN, strong augmentation, a small head with dropout, cross-validation and early stopping.

Hinglish samjho. Chhote data par scratch se mat banao; pretrained model fine-tune karo.

Follow-up. How would you decide how many layers to unfreeze?

Q103 SCENARIO Validation loss is lower than training loss. Is that a problem?

Interview answer. Not always: dropout and augmentation make training loss higher, or the validation set may be easier or leaked. Check the split and compare after evaluating training data in inference mode.

Hinglish samjho. Training mein dropout on hota hai isliye training loss zyada dikhta hai. Split bhi check karo.

Follow-up. How would you confirm there is no leak?

Q104 ADVANCED Why does shuffling the training data each epoch matter?

Interview answer. It avoids correlated batches and consistent ordering, giving better gradient estimates and preventing the model from learning order artifacts.

Hinglish samjho. Order fix ho to model order ko hi seekh sakta hai; shuffle se bachav.

Follow-up. When must you not shuffle?

Q105 INTERMEDIATE What are common ways to monitor and reduce overfitting during training in Keras?

Interview answer. Callbacks such as `EarlyStopping`, `ReduceLROnPlateau` and `ModelCheckpoint`, plus validation data, dropout, `kernel_regularizer` and augmentation layers.

Hinglish samjho. Keras ke callbacks se training khud rukti aur best model save hota hai.

Follow-up. What does `restore_best_weights` do?

Q106 INTERMEDIATE What is label smoothing?

Interview answer. Replacing hard targets (1/0) with softened values (e.g. 0.9/0.1/K) to reduce over-confidence and improve generalisation.

Hinglish samjho. Model ko "100% pakka" bolne se rokta hai.

Follow-up. How does it change the optimal logits?

Q107 INTERMEDIATE What is gradient accumulation?

Interview answer. Summing gradients over several small batches before one update to simulate a larger batch when memory is limited.

Hinglish samjho. Memory kam hai to chhote batches ke gradient jodkar ek baar update.

Follow-up. How must the learning rate and loss scaling be adjusted?

Q108 ADVANCED Why can large-batch training hurt generalisation?

Interview answer. Large batches reduce gradient noise and may converge to sharper minima; learning-rate scaling, warm-up and regularisation are used to compensate.

Hinglish samjho. Shor kam ho to optimizer sharp gaddhe mein phans sakta hai.

Follow-up. How does the number of updates change with batch size?

CHAPTER 13

Interview Q&A · Convolutional Neural Networks

Expect shape and parameter-counting arithmetic, plus explanations of why each component exists.

Q109 BEGINNER What is a CNN and why is it used for images?

Interview answer. A network that applies shared local filters across an image to extract hierarchical features, using far fewer parameters than a dense network and keeping spatial structure.

Hinglish samjho. Ek chhota filter poori image par ghumta hai, to har jagah same pattern pakad leta hai.

Follow-up. Why not flatten the image and use an ANN?

Q110 BEGINNER What is a convolution filter (kernel)?

Interview answer. A small weight matrix (e.g. 3x3) that slides over the input and computes a dot product at each position to produce a feature map.

Hinglish samjho. Filter ek pattern detector hai: edge, corner, texture.

Follow-up. Who chooses the filter values?

Q111 BEGINNER How do grayscale and RGB images differ for a CNN?

Interview answer. Grayscale has one channel ($H \times W \times 1$); RGB has three ($H \times W \times 3$), so each filter has depth 3 and the first convolution combines all channels.

Hinglish samjho. RGB mein filter ki depth 3 hoti hai, ek filter teeno channels ko saath dekhta hai.

Follow-up. How many parameters does one 3x3 filter have on an RGB input?

Q112 BEGINNER What is edge detection with filters?

Interview answer. Filters such as Sobel respond strongly where pixel intensity changes in a given direction, highlighting edges; CNNs learn such filters in early layers.

Hinglish samjho. Jahan brightness achanak badalti hai wahan filter ka output bada hota hai.

Follow-up. What does a horizontal vs vertical edge filter look like?

Q113 BEGINNER What is padding and why use it?

Interview answer. Adding border pixels (usually zeros) so edge pixels are used more and the output size can be preserved ("same" padding).

Hinglish samjho. Padding ke bina har convolution image ko chhota karta jata hai aur kinare ki jaankari kam use hoti hai.

Follow-up. What is the output size with "same" padding and stride 1?

Q114 BEGINNER What is stride?

Interview answer. The number of pixels the filter moves each step; larger strides downsample the output.

Hinglish samjho. Stride 2 matlab filter do pixel kood kar aage badhta hai, output chhota.

Follow-up. How does stride compare with pooling for downsampling?

Q115 INTERMEDIATE Give the output-size formula of a convolution and compute an example.

Interview answer. $\lfloor (n + 2p - f)/s + 1 \rfloor$. For $n = 32$, $f = 5$, $p = 0$, $s = 1$ the output is 28.

Hinglish samjho. Formula yaad rakho: (n + 2p - f) / s + 1.

Follow-up. What if $n = 28$, $f = 3$, $p = 1$, $s = 2$?

Q116 INTERMEDIATE How many parameters does a Conv2D layer have?

Interview answer. $(f \times f \times C_{in} + 1) \times C_{out}$. With 3x3 filters, 3 input channels and 32 filters: $(27 + 1) \cdot 32 = 896$.

Hinglish samjho. Har filter ke f x f x C_in weights aur ek bias, filters ki sankhya se multiply.

Follow-up. Why does the output spatial size not change the parameter count?

Q117 BEGINNER What is pooling and why is it used?

Interview answer. Pooling summarises each local window (max or average) to reduce spatial size, computation and sensitivity to small shifts. It has no learnable parameters.

Hinglish samjho. Image ko chhota karta hai par important cheez bacha leta hai.

Follow-up. Does pooling lose information?

Q118 BEGINNER Compare max pooling and average pooling.

Interview answer. Max pooling keeps the strongest activation (good for detecting presence of a feature); average pooling keeps the mean (smoother). Max is common in hidden layers; global average pooling is common before the classifier.

Hinglish samjho. Max = sabse tez signal, average = ausat.

Follow-up. What is min pooling and when is it used?

Q119 INTERMEDIATE What is global average pooling?

Interview answer. Averaging each whole feature map to one number, which replaces large flatten + dense layers and reduces parameters.

Hinglish samjho. Har feature map ka ek hi average; dense layer ka bojh kam.

Follow-up. How does it help with variable input size?

Q120 BEGINNER What is flattening?

Interview answer. Reshaping the final feature maps into a 1D vector so dense layers can use them.

Hinglish samjho. 2D/3D feature maps ko ek lambi line bana dena.

Follow-up. Where does the flatten size come from in the example 4x4x32?

Q121 INTERMEDIATE What is the receptive field?

Interview answer. The region of the input that influences one output unit; it grows with depth, kernel size and strides.

Hinglish samjho. Ek neuron original image ka kitna hissa "dekh" sakta hai.

Follow-up. How do stacked 3x3 layers compare with one 7x7 layer?

Q122 INTERMEDIATE Why use many small (3x3) filters instead of one large filter?

Interview answer. Stacked small filters reach the same receptive field with fewer parameters and more non-linearity (as in VGG).

Hinglish samjho. Do 3x3 layers = ek 5x5 jitna dekhte hain par kam weights se.

Follow-up. Compare the parameter counts of one 5x5 and two 3x3 layers.

Q123 INTERMEDIATE What is a 1x1 convolution?

Interview answer. A filter that mixes channels at each position, used to change channel count cheaply (bottleneck layers) and add non-linearity.

Hinglish samjho. Spatial nahi, sirf channels ka mixing; channels kam ya zyada karne ke liye.

Follow-up. What does it cost relative to a 3x3 conv?

Q124 INTERMEDIATE What is parameter sharing and translation equivariance?

Interview answer. The same filter weights are applied at every position, so a shifted input produces a shifted feature map (equivariance); pooling adds some invariance.

Hinglish samjho. Bili kahin bhi ho, same filter use pakad leta hai.

Follow-up. Is a CNN invariant to rotation?

Q125 INTERMEDIATE Describe the typical layer order in a CNN.

Interview answer. Conv → (BatchNorm) → ReLU → Pooling, repeated; then Flatten or Global Pooling → Dense → softmax/sigmoid.

Hinglish samjho. Features nikalo, chhota karo, phir classify karo.

Follow-up. Where would you place dropout?

Q126 ADVANCED What are residual connections and why do they matter?

Interview answer. Skip connections add the input of a block to its output, $y = F(x) + x$, letting gradients flow directly and enabling very deep networks (ResNet).

Hinglish samjho. Shortcut raasta gradient ko seedha peeche pahunchata hai.

Follow-up. What does a block do if its weights are zero?

Q127 INTERMEDIATE What is transfer learning in CNNs?

Interview answer. Reusing a CNN pretrained on a large dataset (ImageNet), freezing early layers and fine-tuning later ones for a new task.

Hinglish samjho. Pehle layers general features (edges) seekhti hain, jo har image task mein kaam aate hain.

Follow-up. Which layers do you freeze first?

Q128 INTERMEDIATE **What are common ways to prevent overfitting in CNNs?**

Interview answer. Data augmentation, dropout, weight decay, batch normalisation, early stopping and transfer learning.

Hinglish samjho. Image ko flip, crop, rotate karke data badhao.

Follow-up. Why must augmentation be off during evaluation?

Q129 ADVANCED **What is the difference between a convolution and a transposed convolution?**

Interview answer. A transposed convolution upsamples a feature map (learned upsampling), used in decoders, segmentation and generators.

Hinglish samjho. Convolution image chhoti karta hai, transposed convolution badi.

Follow-up. What is the checkerboard artifact?

Q130 INTERMEDIATE **Compare object classification, detection and segmentation.**

Interview answer. Classification gives one label per image, detection gives boxes with labels, segmentation labels every pixel (semantic or instance).

Hinglish samjho. Kya hai / kahan hai / kis pixel par hai.

Follow-up. Name a model family for each.

Q131 BEGINNER **What is the purpose of the activation function after a convolution?**

Interview answer. To add non-linearity so stacked convolutions can model complex features; ReLU is the usual choice.

Hinglish samjho. Convolution linear hai; activation ke bina kai layers ka matlab kuch nahi.

Follow-up. What happens if you remove it?

Q132 SCENARIO **Your CNN reaches 99% training accuracy but 80% test accuracy on photos from a different camera. What do you do?**

Interview answer. It is overfitting and a domain shift: add augmentation (colour, blur), collect varied data, use transfer learning and regularisation, and evaluate on data from the new domain.

Hinglish samjho. Model ek camera ke style par fit ho gaya; alag-alag cameras ka data aur augmentation chahiye.

Follow-up. How would you measure domain shift?

Q133 SCENARIO How do you reduce the memory use of a CNN on a small device?

Interview answer. Use smaller input sizes, depthwise-separable convolutions, fewer filters, quantisation, pruning or a lightweight architecture.

Hinglish samjho. Model chhota karo ya precision kam karo (quantisation).

Follow-up. What accuracy trade-off do you expect?

Q134 SCENARIO How do you handle a medical imaging task with very few positive cases?

Interview answer. Use transfer learning, strong augmentation, class weights or focal loss, patient-level splits and metrics such as recall, precision and AUC.

Hinglish samjho. Rare bimari ke liye accuracy bekaar; recall dekho aur patient-wise split karo.

Follow-up. Why must the split be by patient, not by image?

Q135 ADVANCED How would you visualise what a CNN looks at?

Interview answer. Use saliency maps, Grad-CAM or feature-map visualisation to see which regions drive the prediction, keeping in mind these are approximate explanations.

Hinglish samjho. Grad-CAM heatmap dikhata hai image ka kaun sa hissa decision ke liye zaroori tha.

Follow-up. What can go wrong when the heatmap highlights the background?

CHAPTER 14

Interview Q&A · RNN, LSTM and GRU

Be able to explain the hidden state, the gates and why long sequences are difficult.

Q136 BEGINNER What is a recurrent neural network?

Interview answer. A network with a feedback loop that carries a hidden state from one time step to the next, so it can process sequences.

Hinglish samjho. Har step par pichhli yaadasht (hidden state) saath leke chalta hai.

Follow-up. What is weight sharing across time?

Q137 BEGINNER Why can't a normal feedforward network handle sequences well?

Interview answer. It has a fixed input size and no memory of order, while sequences vary in length and order carries meaning.

Hinglish samjho. "Dog bites man" aur "man bites dog" alag hain; ANN order nahi samajhta.

Follow-up. How could you still use an ANN for text?

Q138 BEGINNER What is tokenisation?

Interview answer. Splitting text into units (words, subwords or characters) and mapping each to an integer id.

Hinglish samjho. Text ko tukdon mein todkar numbers dena taaki model use padh sake.

Follow-up. Why do modern models prefer subword tokenisation?

Q139 BEGINNER What is an embedding layer?

Interview answer. A trainable lookup table that maps each token id to a dense vector capturing similarity.

Hinglish samjho. Har word ko ek chhota vector, jahan milte-julte words paas hote hain.

Follow-up. Embedding vs one-hot encoding?

Q140 BEGINNER What is the hidden state in an RNN?

Interview answer. A vector summarising the sequence seen so far, computed as $h_t = \tanh(W_x x_t + W_h h_{t-1} + b)$.

Hinglish samjho. Ye RNN ki "yaadasht" hai jo har step par update hoti hai.

Follow-up. What is the shape of h_t?

Q141 INTERMEDIATE What is backpropagation through time (BPTT)?

Interview answer. Unrolling the RNN over time steps and applying backpropagation across them, summing gradients for the shared weights.

Hinglish samjho. RNN ko time mein kholkar ek lamba network maan lo aur usme backprop lagao.

Follow-up. What is truncated BPTT?

Q142 INTERMEDIATE Why do RNNs suffer from vanishing and exploding gradients?

Interview answer. The gradient is multiplied by the recurrent weight matrix (and activation derivative) at every step, so it shrinks or grows exponentially with sequence length.

Hinglish samjho. Ek hi weight matrix baar-baar multiply hota hai; 0.9^50 chhota, 1.1^50 bada.

Follow-up. Which condition on the largest eigenvalue decides which happens?

Q143 BEGINNER What is an LSTM?

Interview answer. A recurrent cell with a cell state and three gates (forget, input, output) that regulate what to keep, write and expose, enabling long-term dependencies.

Hinglish samjho. Conveyor belt (cell state) par gates decide karte hain kya rakhna, kya jodna, kya dikhana.

Follow-up. Why does the additive cell update help gradients?

Q144 INTERMEDIATE Explain each gate of an LSTM.

Interview answer. Forget gate decides what to erase from the cell state, input gate decides how much new candidate information to add and output gate decides how much of the cell state becomes the hidden state.

Hinglish samjho. Forget = bhoolo, Input = naya likho, Output = bahar dikhao.

Follow-up. What is the effect of initialising the forget-gate bias to 1?

Q145 BEGINNER What is a GRU?

Interview answer. A simplified gated cell with an update gate and a reset gate, no separate cell state and fewer parameters than an LSTM.

Hinglish samjho. LSTM ka halka version: kam gates, kam parameters, tez.

Follow-up. Which gate plays the role of both forget and input in an LSTM?

Q146 INTERMEDIATE Compare RNN, LSTM and GRU.

Interview answer. RNN is simplest but forgets quickly; LSTM has the most gates and memory capacity; GRU is lighter and often performs similarly. Choose by data size, sequence length and speed.

Hinglish samjho. Chhoti sequence = RNN/GRU; lambi dependencies = LSTM ya GRU; sabse safe starting point GRU ya LSTM.

Follow-up. Which has more parameters per unit and by how much?

Q147 INTERMEDIATE Why use tanh and sigmoid inside LSTM gates?

Interview answer. Sigmoid outputs (0,1) so it works as a soft on/off switch for gates; tanh outputs (-1,1), keeping candidate values and states centred and bounded.

Hinglish samjho. Gate ko 0-1 chahiye (kitna jaane do), value ko -1 se 1 (sign ke saath).

Follow-up. Why can't a ReLU replace the gate sigmoid?

Q148 INTERMEDIATE What is a bidirectional RNN?

Interview answer. Two RNNs read the sequence forward and backward and their outputs are combined, giving each position both past and future context; not usable for real-time generation.

Hinglish samjho. Sentence ko aage aur peeche dono taraf se padhna.

Follow-up. Why can't you use it for next-word prediction?

Q149 INTERMEDIATE What are many-to-one, one-to-many and many-to-many architectures?

Interview answer. Many-to-one (sentiment, one label per sequence), one-to-many (image captioning), many-to-many aligned (tagging) or unaligned (translation, seq2seq).

Hinglish samjho. Input aur output ki sankhya ke hisaab se alag structure.

Follow-up. Which Keras flag controls many-to-many?

Q150 INTERMEDIATE What is a sequence-to-sequence (encoder-decoder) model?

Interview answer. An encoder compresses the input sequence into a context vector and a decoder generates the output sequence from it; attention was added to avoid the fixed-vector bottleneck.

Hinglish samjho. Ek RNN padhta hai, doosra likhta hai; beech ka vector bottleneck ban jata hai.

Follow-up. How does attention remove that bottleneck?

Q151 INTERMEDIATE What is teacher forcing in RNN training?

Interview answer. Feeding the ground-truth previous token to the decoder at each step during training instead of its own prediction, which speeds convergence but causes exposure bias.

Hinglish samjho. Training mein sahi pichhla answer batate hain, testing mein model apna hi pichhla answer use karta hai.

Follow-up. How does scheduled sampling reduce exposure bias?

Q152 INTERMEDIATE Why do we pad and mask sequences?

Interview answer. Batches need equal length, so shorter sequences are padded; masking tells the layers to ignore padding positions.

Hinglish samjho. Padding sirf jagah bharne ke liye hai, model ko usse seekhna nahi chahiye.

Follow-up. What is `mask_zero=True` in Keras Embedding?

Q153 ADVANCED How does an LSTM mitigate the vanishing gradient mathematically?

Interview answer. The cell state update $c_t = f_t \odot c_{t-1} + i_t \odot \tilde{c}_t$ gives a gradient path $\partial c_t / \partial c_{t-1} = f_t$ that is additive rather than repeated matrix multiplication, so with $f_t \approx 1$ gradients flow almost unchanged.

Hinglish samjho. Cell state par seedhe forget gate se multiply hota hai, bade weight matrix se nahi; isliye gradient bacha rehta hai.

Follow-up. Can LSTMs still have exploding gradients?

Q154 INTERMEDIATE What are the limitations of RNNs, LSTMs and GRUs?

Interview answer. Sequential computation prevents parallelism, long-range dependencies remain hard, training is slow and Transformers usually outperform them on large text tasks.

Hinglish samjho. Ek ek step karke chalna padta hai, isliye slow; Transformers parallel hain.

Follow-up. Where do recurrent models still make sense?

Q155 SCENARIO Your LSTM works on sequences of length 20 but fails on length 500. What do you try?

Interview answer. Use gradient clipping, truncated BPTT or chunking, a GRU/LSTM with better initialisation, attention or a Transformer/convolutional model, and verify masking.

Hinglish samjho. Bahut lambi sequences LSTM ke liye bhi mushkil; attention ya Transformer lagao.

Follow-up. How does memory use grow with sequence length for each model type?

Q156 SCENARIO You forecast demand with an LSTM. How do you split the data?

Interview answer. Use a time-ordered split (train on the past, validate on the future), scale with train statistics only and build windows without crossing the split.

Hinglish samjho. Future ko past predict karne mat do; time-wise split karo.

Follow-up. What baseline would you compare against?

Q157 SCENARIO How would you choose between LSTM and GRU for a small dataset?

Interview answer. Start with GRU for fewer parameters and speed, compare with LSTM on a validation set and keep the simpler model if scores are similar.

Hinglish samjho. Data kam hai to halka GRU pehle try karo.

Follow-up. Which would you pick for very long dependencies?

Q158 ADVANCED Why do stacked recurrent layers need `return_sequences=True` ?

Interview answer. A following recurrent layer expects a full sequence of hidden states, but by default a recurrent layer returns only the last state.

Hinglish samjho. Agle RNN ko har step ka output chahiye, sirf aakhri nahi.

Follow-up. What shape does a Dense layer see after an RNN with `return_sequences=True` ?

CHAPTER 15

Interview Q&A · Transformers and Attention

Be ready to explain Q, K, V in plain words, derive the shapes and compare with RNNs.

Q159 BEGINNER What is the attention mechanism?

Interview answer. A way for a model to weight different input positions when producing each output, using similarity scores turned into weights by softmax.

Hinglish samjho. Har output ke liye model decide karta hai input ke kis hisse par zyada dhyan dena hai.

Follow-up. Why was attention added to seq2seq RNNs?

Q160 BEGINNER What is a Transformer?

Interview answer. A sequence model built from attention and feed-forward layers, without recurrence or convolution, that processes all tokens in parallel.

Hinglish samjho. RNN jaisa ek-ek step nahi, poori sentence ek saath process.

Follow-up. Which paper introduced it?

Q161 BEGINNER What problem do Transformers solve compared with RNNs?

Interview answer. They process sequences in parallel and connect any two positions directly, so training is faster and long-range dependencies are easier to learn.

Hinglish samjho. Parallel training aur seedha long-distance connection.

Follow-up. What new cost do they introduce?

Q162 BEGINNER What are queries, keys and values?

Interview answer. Each token is projected into a query (what it looks for), a key (what it offers) and a value (the information it carries). Query-key similarity decides how much of each value is used.

Hinglish samjho. Query = sawaal, key = label, value = content; match jitna zyada, value utna zyada li jati hai.

Follow-up. Are Q, K, V computed from different inputs in self-attention?

Q163 BEGINNER What is self-attention?

Interview answer. Attention where queries, keys and values all come from the same sequence, so each token gathers information from every other token in that sequence.

Hinglish samjho. Sentence ke har word ko baaki words dekhkar apna matlab samajhna.

Follow-up. How does it differ from cross-attention?

Q164 INTERMEDIATE Write the scaled dot-product attention formula.

Interview answer. $\text{Attention}(Q, K, V) = \text{softmax}(QK^\top / \sqrt{d_k})\, V$.

Hinglish samjho. Scores = Q aur K ka dot product, scale, softmax, phir V ka weighted sum.

Follow-up. What are the shapes of each matrix?

Q165 INTERMEDIATE Why do we divide by $\sqrt{d_k}$?

Interview answer. Dot products grow with the dimension (variance d_k), pushing softmax into saturated regions with tiny gradients; scaling keeps them well-behaved.

Hinglish samjho. Dimension bada ho to dot product bada, softmax flat gradient deta; $\sqrt{d_k}$ se control.

Follow-up. What would happen at $d_k = 512$ without scaling?

Q166 INTERMEDIATE What is multi-head attention and why is it used?

Interview answer. Several attention heads run in parallel on different learned projections and their outputs are concatenated and linearly projected, letting the model capture different relations at once.

Hinglish samjho. Alag-alag heads alag-alag rishte dekhte hain (grammar, meaning, position).

Follow-up. How is d_{model} related to the number of heads?

Q167 INTERMEDIATE What happens after the heads are concatenated?

Interview answer. The concatenated vector (size $h \cdot d_v$) is multiplied by an output projection W_O, returning to d_{model}.

Hinglish samjho. Sab heads ko jodkar ek linear layer se wapas model dimension mein laate hain.

Follow-up. What is the shape of W_O?

Q168 INTERMEDIATE Why do Transformers need positional encoding?

Interview answer. Attention is permutation-invariant, so without position information the model cannot distinguish word order. Position vectors (sinusoidal or learned) are added to embeddings.

Hinglish samjho. Attention ko order ka pata nahi hota; position vector add karke batate hain.

Follow-up. Sinusoidal vs learned positional embeddings?

Q169 ADVANCED Write the sinusoidal positional encoding formula.

Interview answer. $PE_{(pos,2i)} = \sin(pos/10000^{2i/d})$ and $PE_{(pos,2i+1)} = \cos(pos/10000^{2i/d})$.

Hinglish samjho. Har dimension alag frequency ki wave hai, to har position ka alag pattern.

Follow-up. Why does it allow relative-position reasoning?

Q170 INTERMEDIATE What is the encoder-decoder architecture in a Transformer?

Interview answer. The encoder turns the input into contextual representations; the decoder generates the output autoregressively using masked self-attention and cross-attention over the encoder output.

Hinglish samjho. Encoder samajhta hai, decoder likhta hai.

Follow-up. Which models use only the encoder or only the decoder?

Q171 INTERMEDIATE What is masked self-attention in the decoder?

Interview answer. A causal mask sets scores for future positions to $-\infty$ before softmax, so a position can only attend to earlier ones.

Hinglish samjho. Likhte waqt aage ke words dekh nahi sakte, isliye mask lagate hain.

Follow-up. Why is the mask needed during training even though the targets are known?

Q172 INTERMEDIATE What is cross-attention?

Interview answer. Attention where queries come from the decoder and keys and values come from the encoder output, letting the decoder look at the input sequence.

Hinglish samjho. Decoder encoder se poochta hai "input ka kaun sa hissa mere liye kaam ka hai".

Follow-up. Where does it sit in a decoder layer?

Q173 INTERMEDIATE What is the role of the feed-forward network in a Transformer layer?

Interview answer. A position-wise two-layer MLP applied to each token independently, adding non-linearity and capacity after attention (often with an inner size of $4d_{model}$).

Hinglish samjho. Attention tokens ko mila deta hai, FFN har token ko alag se process karta hai.

Follow-up. Does the FFN share weights across positions?

Q174 INTERMEDIATE What are residual connections and layer normalisation doing in each sub-layer?

Interview answer. Residuals add the sub-layer input to its output so gradients flow easily; layer normalisation stabilises activations per token. The original design is Add & Norm after each sub-layer.

Hinglish samjho. Shortcut + normalisation se deep stack stable train hota hai.

Follow-up. Pre-norm vs post-norm?

Q175 INTERMEDIATE Why layer norm and not batch norm in Transformers?

Interview answer. Layer norm works per token and does not depend on batch statistics or sequence length, so it is stable for variable-length sequences and small batches.

Hinglish samjho. Batch ke statistics sentences ke saath ajeeb behave karte hain; layer norm har token ke apne features par kaam karta hai.

Follow-up. What are its learnable parameters?

Q176 BEGINNER What is teacher forcing in Transformer training?

Interview answer. The decoder receives the ground-truth target sequence shifted right as its input, so all positions are trained in parallel with the causal mask.

Hinglish samjho. Training mein sahi pichhle words input dete hain, to ek hi baar mein poori sentence ka loss.

Follow-up. What problem does this create at inference?

Q177 INTERMEDIATE How does inference differ from training for a decoder?

Interview answer. At inference the decoder generates one token at a time using its own previous outputs (autoregressive), often with greedy, beam or sampling decoding and cached keys/values.

Hinglish samjho. Test time par pichhla token model ka apna output hota hai, isliye slow aur ek-ek karke.

Follow-up. What is a KV cache?

Q178 INTERMEDIATE What are contextual embeddings and how do they differ from Word2Vec?

Interview answer. Contextual embeddings depend on the surrounding sentence (the same word gets different vectors); traditional embeddings are a fixed vector per word.

Hinglish samjho. "Bank" river ke paas alag vector, paise ke saath alag.

Follow-up. Which models produce contextual embeddings?

Q179 INTERMEDIATE What is the computational complexity of self-attention?

Interview answer. $O(n^2 d)$ time and $O(n^2)$ memory in sequence length n, versus $O(n\,d^2)$ and $O(n)$ steps for an RNN.

Hinglish samjho. Sequence double karo to attention ka kharcha chaar guna.

Follow-up. How do efficient attention variants reduce it?

Q180 ADVANCED How many parameters are in one multi-head attention block?

Interview answer. Four projection matrices of size $d_{model} \times d_{model}$ (for Q, K, V and output), so about $4d_{model}^2$ plus biases, independent of sequence length.

Hinglish samjho. Q, K, V aur output ke 4 weight matrices.

Follow-up. How many in the FFN with inner size $4d_{model}$?

Q181 ADVANCED What happens if the number of heads is increased while d_{model} stays fixed?

Interview answer. Each head gets a smaller dimension d_{model}/h, keeping total compute similar; too many heads can hurt because each head becomes too small to compute useful compatibility.

Hinglish samjho. Zyada heads matlab har head ka size chhota.

Follow-up. How can attention heads be pruned?

Q182 INTERMEDIATE Compare BERT-style and GPT-style models.

Interview answer. BERT is an encoder trained with masked-token prediction for understanding tasks; GPT is a decoder trained for next-token prediction for generation.

Hinglish samjho. BERT dono taraf dekhta hai (samajhne ke liye), GPT sirf peeche dekhta hai (likhne ke liye).

Follow-up. Which would you use for classification and which for chat?

Q183 INTERMEDIATE **Why are Transformers faster to train than RNNs?**

Interview answer. Computation over all positions is done in parallel with matrix multiplications that suit GPUs, while RNNs must process time steps sequentially.

Hinglish samjho. GPU ko ek saath bahut saara matrix multiply pasand hai.

Follow-up. Is inference faster too?

Q184 ADVANCED **What is the difference between absolute and relative positional encodings?**

Interview answer. Absolute encodings add a vector per position; relative schemes (such as rotary or relative biases) encode the distance between tokens inside attention and extrapolate better to longer inputs.

Hinglish samjho. Absolute = "main 5th hoon", relative = "tum mujhse 2 door ho".

Follow-up. Why do longer inputs than training length cause problems?

Q185 SCENARIO **You must classify long documents of 10,000 tokens with a Transformer. What do you do?**

Interview answer. Chunk the text and aggregate predictions, use a long-context or sparse-attention model, retrieve relevant passages first or summarise hierarchically.

Hinglish samjho. Poora document ek saath nahi jayega; tukdon mein baanto ya long-context model lo.

Follow-up. What are the trade-offs of chunking?

Q186 SCENARIO **How would you fine-tune a pretrained Transformer on a small labelled dataset?**

Interview answer. Add a task head, use a small learning rate, freeze or gradually unfreeze layers, apply early stopping and evaluate with a held-out set.

Hinglish samjho. Pretrained model ko chhote learning rate se apne data par thoda sa train karo.

Follow-up. What signs tell you that you are overfitting?

Q187 SCENARIO **Your translation model generates repeated phrases. What could help?**

Interview answer. Use beam search with a repetition or length penalty, check training data quality, increase training or use coverage and sampling controls such as top-p.

Hinglish samjho. Decoding settings aur repetition penalty se repeat kam hota hai.

Follow-up. Beam search vs sampling?

Q188 ADVANCED **Why can attention weights not be treated as a perfect explanation?**

Interview answer. Different attention patterns can give the same output and attention is one of several pathways (residual, FFN), so it indicates correlation, not guaranteed causation.

Hinglish samjho. Attention heatmap ek ishara hai, pakka saboot nahi.

Follow-up. Which methods give better attribution?

Q189 INTERMEDIATE **What is the shape flow in self-attention for a batch of B sequences of length n with model size d?**

Interview answer. Input (B, n, d) → Q,K,V (B, n, d) (split to $(B, h, n, d/h)$) → scores (B, h, n, n) → weights (B, h, n, n) → output (B, n, d).

Hinglish samjho. Scores ka matrix n x n hota hai, isliye lambi sequence mahngi padti hai.

Follow-up. Which dimension does softmax run over?

CHAPTER 16

Interview Q&A · Coding, Debugging and Scenario Practice

Short coding and debugging prompts. Answer with the idea first, then the code line or fix. Tools used: Keras 3 / TensorFlow 2.

Q190 BEGINNER **Write the Keras code for a small binary classifier on 20 features.**

Interview answer. `Sequential([Input((20,)), Dense(16,"relu"), Dense(1,"sigmoid")])`, then `compile("adam","binary_crossentropy",metrics=["accuracy"])` and `fit(X,y,epochs=20,validation_split=0.2)`.

Hinglish samjho. Input shape, hidden ReLU layer, sigmoid output; compile mein loss aur optimizer; phir fit.

Follow-up. How would you add early stopping?

Q191 BEGINNER **How do you compute the number of trainable parameters of `Dense(64)` on a 100-feature input?**

Interview answer. $100 \times 64 + 64 = 6464$ (weights plus biases); `model.count_params()` confirms it.

Hinglish samjho. inputs x units + units (bias).

Follow-up. How many for a following `Dense(10)` ?

Q192 INTERMEDIATE **What is the bug: `Dense(10, activation="softmax")` with `loss="categorical_crossentropy"` and integer labels?**

Interview answer. Integer labels need `sparse_categorical_crossentropy`, or the labels must be one-hot encoded with `to_categorical`.

Hinglish samjho. Label ka format loss se match hona chahiye: integer = sparse, one-hot = categorical.

Follow-up. What error or silent problem appears if you ignore this?

Q193 INTERMEDIATE **The model uses `Dense(1, activation="sigmoid")` with `loss=BinaryCrossentropy(from_logits=True)`. What is wrong?**

Interview answer. The sigmoid is applied twice; remove the sigmoid from the model or set `from_logits=False`.

Hinglish samjho. Logits flag ka matlab model ke output mein pehle se sigmoid nahi hona chahiye.

Follow-up. How does it show up in the training curves?

Q194 INTERMEDIATE — Why does validation accuracy look worse after you added `Dropout` and training accuracy is lower than validation?

Interview answer. Dropout is active in training only, so reported training metrics are noisier and lower; validation runs in inference mode.

Hinglish samjho. Training mein neurons band hain isliye training score kam dikhta hai.

Follow-up. How would you measure the training accuracy fairly?

Q195 INTERMEDIATE — Fix: the text model fails with `ValueError` when stacking two LSTM layers.

Interview answer. The first LSTM needs `return_sequences=True` so the second receives a 3D tensor `(batch, time, features)`.

Hinglish samjho. Pehla LSTM poori sequence return kare, tabhi doosra use le sakta hai.

Follow-up. What shape does the last LSTM return by default?

Q196 INTERMEDIATE — Fix: the CNN reports `Negative dimension size` after several layers.

Interview answer. The feature map became smaller than the kernel: use `padding="same"`, fewer poolings or a larger input size.

Hinglish samjho. Image itni chhoti ho gayi ki filter fit nahi hota; padding same lagao.

Follow-up. Compute the size after Conv(3x3, valid) then MaxPool(2) on 8x8.

Q197 INTERMEDIATE — Fix: accuracy is 10% on a 10-class problem and loss is stuck near 2.30.

Interview answer. $\ln 10 \approx 2.30$ means random guessing: check labels versus inputs alignment, pixel scaling, learning rate, shuffling and that the output uses softmax with the right loss.

Hinglish samjho. 2.30 = sirf andaza; data aur labels ka mel check karo, inputs scale karo.

Follow-up. What loss do you expect from random guessing on 5 classes?

Q198 INTERMEDIATE — Write a custom training step with `GradientTape`.

Interview answer. `with tf.GradientTape() as t: loss = loss_fn(y, model(x, training=True))` then `grads = t.gradient(loss, model.trainable_variables)` and `opt.apply_gradients(zip(grads, model.trainable_variables))`.

Hinglish samjho. Tape forward pass record karta hai, phir gradient nikalkar optimizer update karta hai.

Follow-up. Why pass `training=True` ?

Q199 INTERMEDIATE **How do you save and load a Keras model correctly?**

Interview answer. `model.save("m.keras")` and `keras.models.load_model("m.keras")`; include the preprocessing (scaler or layers) used in training.

Hinglish samjho. Model ke saath preprocessing bhi save karo, warna prediction galat hogi.

Follow-up. What goes wrong if the scaler is not saved?

Q200 INTERMEDIATE **How do you apply class weights in Keras?**

Interview answer. Pass `class_weight={0:1.0, 1:5.0}` to `fit`, computed from class frequencies, so errors on the rare class cost more.

Hinglish samjho. Rare class ki galti ko zyada weight.

Follow-up. What else would you use besides class weights?

Q201 ADVANCED **Implement scaled dot-product attention in NumPy.**

Interview answer. `scores = Q @ K.T / np.sqrt(d_k)`; `w = softmax(scores, axis=-1)`; `out = w @ V`, subtracting the row max before `exp` for stability.

Hinglish samjho. Score, scale, softmax, V se weighted sum.

Follow-up. How would you add a causal mask?

Q202 ADVANCED **Implement a causal mask for a sequence length n.**

Interview answer. `mask = np.triu(np.ones((n,n)), k=1).astype(bool)`; set `scores[mask] = -1e9` before softmax so future positions get ~0 weight.

Hinglish samjho. Upper triangle ko bahut negative bana do.

Follow-up. Why use $-\infty$ or a large negative number rather than 0?

Q203 INTERMEDIATE **Write the Keras code for early stopping and checkpointing.**

Interview answer. `callbacks=[EarlyStopping(patience=5, restore_best_weights=True), ModelCheckpoint("best.keras", save_best_only=True)]` passed to `fit`.

Hinglish samjho. Validation loss na sudhre to ruk jao aur best weights rakho.

Follow-up. What should `monitor` be set to?

Q204 SCENARIO **A model trains fine, but predictions in production are poor. List what you check.**

Interview answer. Training-serving skew: preprocessing differences, label or feature drift, different tokenizer or image resizing, batch-norm/dropout mode and data schema changes.

Hinglish samjho. Production mein wahi scaling, tokenizer aur image size honi chahiye jo training mein thi.

Follow-up. How would you monitor drift after deployment?

Q205 SCENARIO **You get an out-of-memory error when training. What are the options?**

Interview answer. Reduce the batch size or input size, use mixed precision, gradient accumulation or checkpointing, a smaller model and generators/ `tf.data` pipelines.

Hinglish samjho. Batch chhota karo, mixed precision lagao, data stream karo.

Follow-up. What does mixed precision trade off?

Q206 SCENARIO **Two runs with the same code give different accuracies. Why and what do you do?**

Interview answer. Random initialisation, shuffling and non-deterministic GPU ops; set seeds (`keras.utils.set_random_seed`), report mean and standard deviation over several runs.

Hinglish samjho. Seed fix karo aur 3-5 runs ka average batao.

Follow-up. What does enabling op determinism cost?

Q207 SCENARIO **Design a quick plan to build a sentiment classifier from 20,000 reviews.**

Interview answer. Clean and split the data, build a TF-IDF + logistic regression baseline, then a small embedding+GRU/LSTM or fine-tune a pretrained Transformer, compare with F1 on a held-out test set and error-analyse.

Hinglish samjho. Pehle simple baseline banao; phir deep model aaye to dekho fayda hua ya nahi.

Follow-up. When is the baseline enough?

Q208 SCENARIO **How would you detect and handle a mislabelled-data problem?**

Interview answer. Look at the highest-loss training samples, check agreement between models, relabel or remove noisy ones and use robust losses or label smoothing.

Hinglish samjho. Jin samples par loss sabse zyada hai unhe khud dekho.

Follow-up. How much label noise can a network tolerate?

Q209 ADVANCED **Explain how you would unit-test a neural network training pipeline.**

Interview answer. Check output shapes, that loss decreases on a tiny dataset, that gradients are non-zero for all trainable variables, that the model can overfit one batch and that saving/loading gives identical predictions.

Hinglish samjho. Chhoti batch par overfit karwa ke dekho; gradients aur shapes check karo.

Follow-up. Which of these catches a missing activation?

Q210 SCENARIO **How would you explain a model's decision to a non-technical stakeholder?**

Interview answer. Use simple feature-importance or example-based explanations, show typical correct and wrong cases, state limits and uncertainty and avoid overclaiming what a heatmap proves.

Hinglish samjho. Seedhe shabdon mein batao kaunse factors zyada zaroori the aur kahan model galat ho sakta hai.

Follow-up. What would you say if the explanation contradicts domain knowledge?

CHAPTER 17

Mock Interview Practice

1. How to Run a Mock Round

Hinglish mein samjho

Kisi dost ko ya khud ko timer ke saath sawaal poochho. Pehle 30 second mein seedha jawab, phir 1 minute mein example ya formula, aur aakhir mein follow-up. Jab atak jao to bolo "mujhe jahan tak pata hai..." aur apni soch batao, interviewer wahi dekhte hain.

Round	Time	Take questions from	Goal
1. Warm-up	15 min	Perceptron and ANN, Activations, Losses (beginner)	Crisp definitions in 30 seconds
2. Core	25 min	Optimizers, CNN, RNN/LSTM/GRU (intermediate)	Formula plus intuition plus example
3. Depth	25 min	Transformers, Training and regularisation (advanced)	Reason about trade-offs
4. Scenario and coding	25 min	Coding, Debugging and Scenario Practice	Debug, design, decide

Scoring rubric (0-2 each): correct definition, correct formula or shape, a concrete example, a trade-off or limitation, clear communication. Revisit any question that scores under 6 of 10.

2. Model Answers in the Interview Format

Q: Explain vanishing gradients and how LSTM helps (about 90 seconds)

Say: "In deep or recurrent nets the gradient is a product of many derivatives. If each is below 1 (sigmoid's derivative is at most 0.25) the product shrinks exponentially, so early layers or early time steps barely learn. An LSTM keeps a cell state updated *additively*, $c_t = f_t c_{t-1} + i_t \tilde{c}_t$, so the gradient along the cell path is multiplied only by the forget gate, which can stay near 1. For feedforward nets I would also use ReLU, He initialisation, batch norm or residual connections." **Likely follow-up:** "Can LSTMs explode?" Yes, so I clip gradients.

Q: Walk me through self-attention (about 2 minutes)

Say: "Each token is projected to a query, key and value. Scores are $QK^\top/\sqrt{d_k}$, softmax turns each row into weights, and the output is the weighted sum of values, so each token becomes a context-aware mixture of all tokens. Several heads do this in parallel; I concatenate them and apply W_O. Positional encoding supplies order. Cost is $O(n^2 d)$." **Follow-up:** "Why divide by $\sqrt{d_k}$?" To stop large dot products saturating the softmax.

3. Debugging Clinic

Symptom	Likely cause	Fix
Loss stays at $\ln K$ (e.g. 0.693 binary, 2.30 for 10 classes)	Not learning: wrong labels, no scaling, tiny or huge learning rate	Overfit 20 samples first; scale inputs; tune η
Loss becomes NaN	Exploding gradients, $\log 0$, bad inputs	Lower η, `clipnorm`, clip probabilities, check data
Train good, validation poor	Overfitting or leakage	Dropout, augmentation, early stopping, check split
Both poor	Underfitting or bug	Bigger model, train longer, check pipeline
Validation better than train	Dropout/augmentation on in training, easy validation set	Compare in inference mode, check leakage
Accuracy fine, one class never predicted	Class imbalance	Class weights, focal loss, threshold, per-class metrics

Three typical bugs with their fixes:

```
BUG 1: model.add(Dense(10, activation="softmax"));  loss="categorical_crossentropy"  with integer labels
FIX  : use loss="sparse_categorical_crossentropy"  (or one-hot the labels with to_categorical)

BUG 2: LSTM(32) followed by LSTM(32)
FIX  : the first layer needs return_sequences=True

BUG 3: scaler.fit_transform(X_test)
FIX  : scaler.fit(X_train); X_test = scaler.transform(X_test)   # never fit on test data
```

4. Last-Minute Checklist

Before the interview

- Write the forward/backward equations of one neuron, and Adam's update, from memory.
- Recall: output-size formula $\lfloor (n + 2p - f)/s + 1 \rfloor$, conv parameters $(f^2 C_{in} + 1) C_{out}$, attention formula, LSTM gate roles.
- Know the output layer + loss pairs for binary, multi-class, multi-label and regression.
- Prepare one project story: problem, data, model, metric, what failed, what you changed.
- **Memory tricks:** "FLBU" (forward, loss, backward, update); "Sparse = integer, Categorical = one-hot"; "LSTM: F-I-O, GRU: U-R"; "Q asks, K matches, V gives".
- If you do not know an answer: state what you know, reason aloud, and say how you would verify it.

PART D

Reference

18. Final Cheat Sheet

19. References and Further Reading

CHAPTER 18

Final Cheat Sheet

Ideas in Plain Words

Topic	Remember this
Learning loop	Guess (forward) → grade (loss) → blame (backprop) → fix (update)
Weight update	New weight = old weight − learning rate × gradient
Chain rule	Blame is passed back by multiplying the effect of each step
Sigmoid	Squeezes to 0..1; slope at most 0.25, so it causes vanishing gradients
ReLU	Negative becomes 0, positive passes; the default hidden activation
Softmax	Scores become probabilities that add to 1
MSE / MAE / Huber	Square of error / plain size of error / mix of both
Cross-entropy	Big penalty when confident and wrong: loss = −ln(probability of the true class)
Momentum	Keep part of the old speed
AdaGrad	Smaller steps for weights updated often
Adam	Momentum + adaptive steps; start with learning rate 0.001
CNN output size	(image − filter + 2 × padding) ÷ stride + 1
CNN weights	(filter × filter × input channels + 1) × number of filters
RNN	New memory = function(new word, old memory)
LSTM	Forget gate erases, input gate writes, output gate speaks
GRU	Update gate and reset gate
Attention	softmax(Q·K ÷ √size) × V

Quick Choices

Task	Last layer	Loss
Yes/no	1 sigmoid	Binary cross-entropy
One class out of many (one-hot labels)	softmax	Categorical cross-entropy
One class out of many (integer labels)	softmax	Sparse categorical cross-entropy
Many tags	sigmoid on each	Binary cross-entropy
A number	linear	MSE, MAE or Huber

Data	Use	Why
Tables	ANN (or boosting)	No shape or order to exploit
Images	CNN	Filters find patterns anywhere
Sequences (small)	LSTM / GRU	Memory of the past
Text, long context	Transformer	Attention to every word, parallel training

Common default	Choice
Hidden activation	ReLU
Optimizer	Adam (0.001)
Starting weights	He for ReLU, Glorot for tanh/sigmoid
Stop overfitting	Dropout, augmentation, early stopping, weight decay
Normalisation	Batch norm (CNN), layer norm (Transformer)

Last-Minute Reminders

Remember

- **Vanishing gradients:** sigmoid/tanh flatten out; fix with ReLU, good starting weights, batch norm, shortcuts, LSTM/GRU. **Exploding:** clip the gradient.
- **Labels:** sparse = integer, categorical = one-hot. If the loss says `from_logits=True`, the model must **not** end with softmax or sigmoid.
- **Training vs testing:** dropout is off at test time; decoders use teacher forcing in training and generate word by word at test time.
- **Never** fit scalers on test data; touch the test set only once.
- Always answer with: meaning, how it works, tiny example, one limit.

CHAPTER 19

References and Further Reading

How these sources were checked

Links were found with web search while building this handbook (November 2026). ✔ = **the exact URL appeared in search results** for the stated topic. ○ = **cited by name or appears only inside another page's reference list; link not individually confirmed.** Public interview pages were used to see which *themes* are commonly covered. All practice questions and explanations here are written in original wording and **none is attributed to a company**; no question-frequency claims are made.

Official documentation

Topic	Link	Status
Keras layers API (Dense, Conv, recurrent, attention, normalisation)	keras.io/api/layers	✔
Keras `Dense` layer	keras.io/api/layers/core_layers/dense	✔
Keras `MultiHeadAttention`	keras.io/api/layers/attention_layers/multi_head_attention	✔
Keras Functional API guide (training, evaluation, losses, optimizers)	keras.io/guides/functional_api	✔
TensorFlow `tf.keras.layers` module	tensorflow.org/api_docs/python/tf/keras/layers	✔
TensorFlow `MultiHeadAttention`	tensorflow.org/.../MultiHeadAttention	✔
TensorFlow Functional API guide	tensorflow.org/guide/keras/functional_api	✔

Interview-preparation resources

Resource	Link	Status and honest note
GeeksforGeeks: Deep Learning Interview Questions	geeksforgeeks.org/deep-learning/deep-learning-interview-questions	✔ Large Q&A list covering initialisation, optimizers, dropout, CNN receptive field, RNN/LSTM/GRU, layer normalisation
GeeksforGeeks: AI interview questions (neural network, CNN vs RNN, Transformer)	geeksforgeeks.org/.../artificial-intelligenceai-interview-questions-and-answers	✔
GeeksforGeeks: AI / ML / DS interview series	geeksforgeeks.org/interview-experiences/ai-ml-ds-interview	✔ Hub page listing common deep learning themes
GeeksforGeeks: NLP interview questions (RNN, BPTT, seq2seq)	geeksforgeeks.org/nlp/nlp-interview-questions	✔
GeeksforGeeks: Computer vision interview questions	geeksforgeeks.org/computer-vision-interview-questions	✔
GeeksforGeeks quizzes: Neural Networks; Intro to CNN; Intro to Deep Learning	neural-networks-quiz, intro to CNN, intro to DL	✔ Multiple-choice self-tests
Naukri Code360 (Coding Ninjas): interview questions library	naukri.com/code360/library/interview-questions	✔ General library; includes basic "What is Deep Learning?" style entries
Naukri Code360: a candidate-submitted Deep Learning Engineer interview experience	naukri.com/code360/interview-experiences/dubverse/...	✔ One person's account (Mar 2020) mentioning topics such as loss functions, backpropagation mathematics and attention. Treated only as an example of themes, **not** as evidence of how often anything is asked
Shiksha / Naukri Learning: Deep Learning interview questions	naukri.com/learning/articles/deep-learning-interview-questions-and-answers	✔
LeetCode community post on AI/ML problem sets	leetcode.com/discuss/.../LeetCode-Problem-sets-to-prepare-for-AIML-interview	✔ A user post (not official) noting that LeetCode has no dedicated AI/ML problem set. Use LeetCode for Python and algorithm rounds, not deep learning theory
Educative: ML coding interview preparation	educative.io/blog/machine-learning-coding-interview-questions	✔

Papers

Paper	Link	Status
Vaswani et al., *Attention Is All You Need* (2017)	arxiv.org/abs/1706.03762	✔
Kingma and Ba, *Adam: A Method for Stochastic Optimization*	arxiv.org/abs/1412.6980	✔
Hochreiter and Schmidhuber, *Long Short-Term Memory* (1997)	cited by name (TensorFlow's LSTM docs attribute the layer to Hochreiter 1997)	○
Cho et al., *Learning Phrase Representations using RNN Encoder-Decoder* (2014), the GRU paper	arXiv:1406.1078, seen inside another paper's reference list	○

Honest limitations

- Interview questions come from public themes; real interviews vary by company and role, and no source here proves what any employer asks.
- The code uses tiny datasets so it runs offline in seconds. Accuracy numbers (for example 100% on the toy Transformer task) show the mechanics, not realistic performance.
- Computer vision beyond the CNN basics, generative models, reinforcement learning, large language model deployment and MLOps are outside this handbook.